

ACHIEVE!

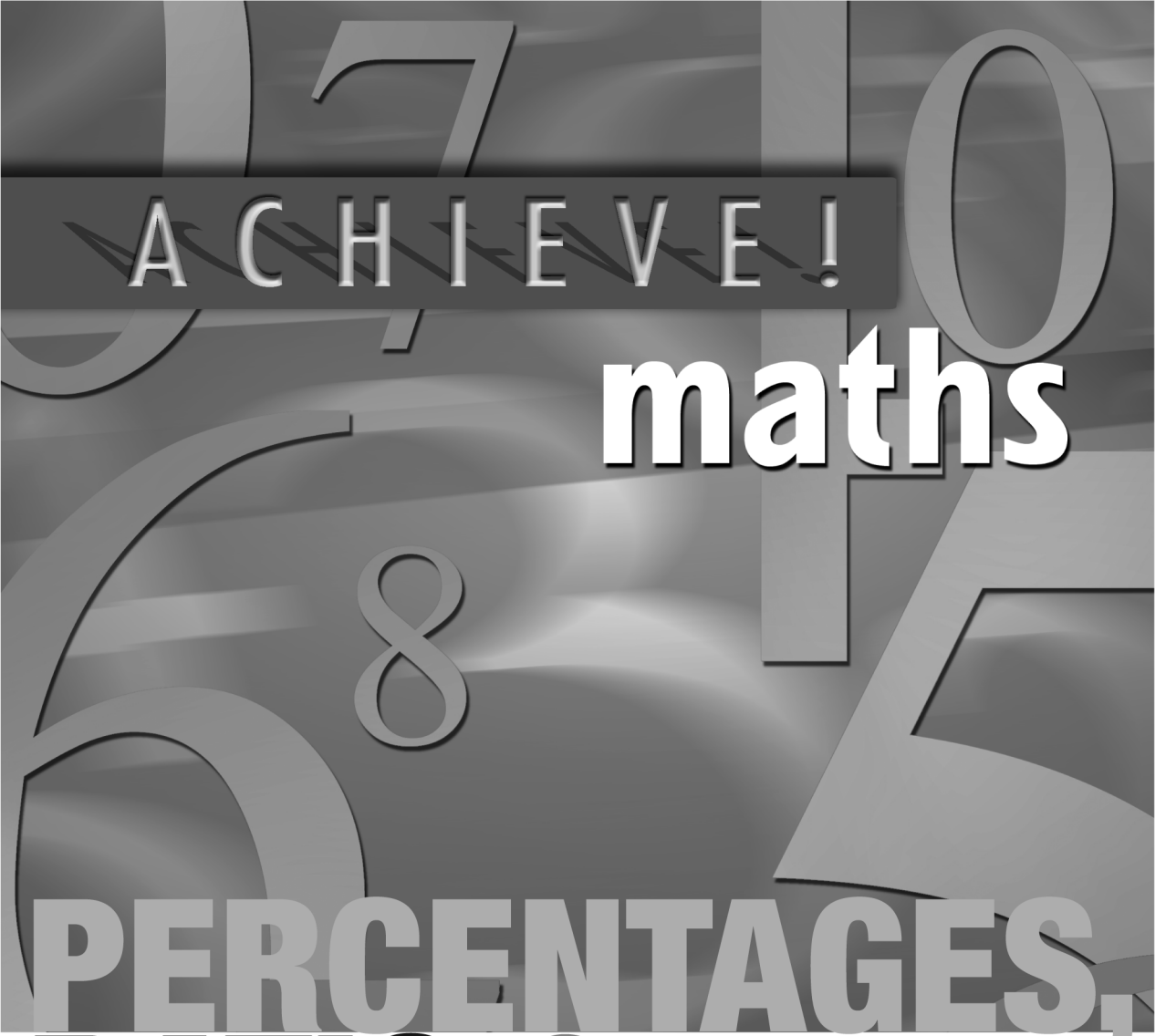


maths

**PERCENTAGES,
RATIO &
PLANE SHAPES**

Paul Steenson



The background of the cover is a grayscale abstract design. It features large, stylized numbers and mathematical symbols. At the top, a large '7' is on the left and a large '0' is on the right. In the center, there is a large '8'. Below the '8', there is a large '4'. The word 'ACHIEVE!' is written in a bold, sans-serif font across the top. The word 'maths' is written in a large, white, sans-serif font with a black outline, positioned below 'ACHIEVE!'. The words 'PERCENTAGES,' 'RATIO &' and 'PLANE SHAPES' are written in large, bold, sans-serif fonts, stacked vertically. The overall design is modern and geometric.

ACHIEVE!

maths

PERCENTAGES,
RATIO &
PLANE SHAPES

Paul Steenson

This title was commissioned by Jennifer Gough, project management was by Piper Print Media and Gail McManus Graphics designed the cover.

Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has also been a co-author of textbooks and written a number of titles in the Instant Lessons series.

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Contents



Introduction	4
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Unit 1: Percentages

Teacher's notes	5
1.1 Fractions of 100	7
1.2 Fractions and percentages	9
1.3 Decimals and percentages	10
1.4 What percentage is that?	11
1.5 Fractions, decimals and percentages	12
1.6 Equivalent fractions, decimals and percentages	14
1.7 Calculating a percentage A	15
1.8 Calculating a percentage B	16

Unit 2: Using percentages

Teacher's notes	18
2.1 Percentage increase	20
2.2 Percentage decrease	22
2.3 Discounts	24
2.4 Percentage composition	26
2.5 Percentage profit	28
2.6 Percentage loss	30

Unit 3: Ratio

Teacher's notes	32
3.1 What is a ratio?	34
3.2 Equivalent ratios	36
3.3 Using equivalent ratios	37
3.4 Simplifying ratios	38
3.5 Increase in a ratio	40
3.6 Decrease in a ratio	41
3.7 Dividing in a ratio	42
3.8 Dividing into 3 parts	44

Unit 4: Plane shapes

Teacher's notes	45
4.1 Polygons A	46
4.2 Polygons B	47
4.3 Triangles	48
4.4 Quadrilaterals	50
4.5 Line symmetry	52

Unit 5: Using drawing instruments

Teacher's notes	53
5.1 Parallel lines	54
5.2 Perpendicular lines	55
5.3 Using a protractor	56
5.4 Measuring angles	57
5.5 Drawing angles	58
5.6 Using compasses	59

Unit 6: Constructing plane shapes

Teacher's notes	60
6.1 Constructing rectangles and squares	61
6.2 Constructing parallelograms	62
6.3 Constructing triangles	63
6.4 Constructing regular polygons	64

Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into six units, each of which focuses on a topic or set of skills to do with percentages, ratio or plane shapes. The CD contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. The activity sheet documents can be easily edited to suit the needs of students and the curriculum requirements of your state.

Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit.

The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the activity sheets.

Prior knowledge – This refers to the skills and knowledge that are required for students to complete the tasks.

Background – This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity sheets and answers –

A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

Extension ideas – Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

Curriculum links – The activity sheets will enable students to achieve outcomes required for the following State levels or stages:

- New South Wales – Stages 4 and 5
- Victoria – Levels 5 and 6
- Queensland – Levels 4, 5 and 6



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Teacher's notes

OBJECTIVES

- Understand the concept of a percentage as a fraction of 100.
- Convert fractions to percentages and percentages to fractions.
- Convert decimals to percentages and percentages to decimals.
- Recognise and recall common equivalent fractions, decimals and percentages.
- Express one quantity as a percentage of another quantity.
- Calculate a percentage of a quantity.
- Solve simple problems involving percentages.

PRIOR KNOWLEDGE

The following activity sheets have been developed for students to complete without using a calculator. Students should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or pen and paper methods.

BACKGROUND

Percentages are commonly used in many areas of our daily life. The word 'cent' comes from the Latin word *centum* meaning 'one hundred'. *Percent* means 'out of one hundred'.

STARTER ACTIVITY

Lead a discussion about percentages. Ask students what they know about percentage, and where in the real world they see percentages (e.g. test scores, sales catalogues, financial institutions). Use a mind map to record their ideas on a large sheet of paper, to review at the end of the topic.

ACTIVITY SHEETS

1.1 Fractions of 100

This activity sheet develops the concept of a percentage as a fraction of one hundred. The term **denominator** may need explanation for some students. Note that it is not intended that students simplify their answers in question 6.

Answers

1 a 85% b 60% c 74%

2 a $\frac{80}{100} = 80\%$ b $\frac{95}{100} = 95\%$ c $\frac{30}{100} = 30\%$

d $\frac{75}{100} = 75\%$

3 a 40 b $\frac{40}{100}$ or $\frac{2}{5}$ c 40% d 60%

4 a 65% b 58%

5 b 15% c 60% d 30% e 90% f 75%

g 42% h 86% i 5%

6 b $\frac{50}{100}$ c $\frac{35}{100}$ d $\frac{85}{100}$ e $\frac{60}{100}$ f $\frac{15}{100}$

g $\frac{4}{100}$ h $\frac{92}{100}$ i $\frac{38}{100}$

7 35%

8 80%

1.2 Fractions and percentages

This activity sheet develops the skill of converting a fraction to a percentage.

Answers

1 a 80% b 75% c 40% d 30% e 70%

f 90% g 55% h 45% i 85% j $12\frac{1}{2}\%$

k $37\frac{1}{2}\%$ l $33\frac{1}{3}\%$ m $66\frac{2}{3}\%$ n $83\frac{1}{3}\%$

1.3 Decimals and percentages

This activity sheet develops the skill of converting a decimal to a percentage.

Answers

1 a 14% b 35% c 72% d 83% e 6%

f 85% g 40% h 50% i 30% j 12.5%

k 62.5% l 74.5% m 85.9% n 10.6%

o 4.3% p 48% q 5% r 11.5% s 125%

t 165% u 105% v 150% w 190%

x 250%

1.4 What percentage is that?

How many times have students asked this question when you hand back their test results as a mark out of 20? This activity sheet develops the skill of expressing one quantity as a percentage of another quantity.

Answers

1 75%

2 40%

3 72%

4 30%

5 25%

6 19%

7 62.5%

8 12%

9 84%

Teacher's notes

CONTINUED

1.5 Fractions, decimals and percentages

This activity sheet provides students with further practice at converting between fractions, decimals and percentages.

Answers

- 1 a 27% b 93% c 11% d 70% e 40%
 f 75% g $12\frac{1}{2}\%$ h $87\frac{1}{2}\%$ i $66\frac{2}{3}\%$
 2 a 62% b 37% c 25% d 80% e 50%
 f 20% g 4% h 12.5% i 87.5%
 3 a 0.59 b 0.75 c 0.1 d 0.35 e 0.2
 f 0.5 g 0.63 h 0.8 i 0.05 j 0.125
 k 0.675 l 0.095 m 1.15 n 1.25 o 1.3
 4 a $\frac{1}{10}$ b $\frac{1}{4}$ c $\frac{1}{5}$ d $\frac{2}{5}$ e $\frac{9}{10}$ f $\frac{3}{4}$ g $\frac{9}{25}$
 h $\frac{3}{20}$ i $\frac{17}{20}$ j $\frac{1}{20}$ k $\frac{19}{25}$ l $\frac{23}{25}$ m $\frac{1}{8}$ n $\frac{3}{8}$
 o $\frac{2}{3}$ p $1\frac{1}{5}$ q $1\frac{1}{2}$ r $1\frac{3}{4}$

1.6 Equivalent fractions, decimals and percentages

In this activity sheet, students complete a table of equivalent decimals and percentages for common fractions (denominators 2, 3, 4, 5, 8 and 10).

It is recommended that students learn the commonly used equivalencies in this table for later recall. This will assist them in choosing the appropriate equivalent form for mental computation. For example, 25% of \$40 is $\frac{1}{4}$ of \$40.

Answers

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.3	$33\frac{1}{3}\%$ or $33.\dot{3}\%$
$\frac{2}{3}$	0.6	$66\frac{2}{3}\%$ or $66.\dot{6}\%$
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{8}$	0.125	$12\frac{1}{2}\%$ or 12.5%
$\frac{3}{8}$	0.375	$37\frac{1}{2}\%$ or 37.5%
$\frac{5}{8}$	0.625	$62\frac{1}{2}\%$ or 62.5%

$\frac{7}{8}$	0.875	$87\frac{1}{2}\%$ or 87.5%
$\frac{1}{10}$	0.1	10%
$\frac{3}{10}$	0.3	30%
$\frac{7}{10}$	0.7	70%
$\frac{9}{10}$	0.9	90%

1.7 Calculating a percentage A

This activity sheet develops the skill of calculating a percentage of a quantity, using the equivalent common fraction. Students need to be able to calculate a fraction of a quantity before attempting this activity sheet.

Answers

- 1 a \$30 b \$36 c 75 km
 2 a \$20 b \$14 c 44 kg
 3 a \$12 b \$6.50 c 7 tonnes
 4 a \$40 b \$19 c 11 metres d 26 kg

1.8 Calculating a percentage B

This activity sheet shows students a general method for calculating a percentage of a quantity. Encourage students to write at least one line of working as shown in the example. Most students will need to use their calculator to complete the second page.

Answers

- 1 a \$24 b \$72 c \$70 d \$210 e \$64
 f \$10 g \$95 h 45 m i 140 kg j 840 km
 k 360 g l 2 880 kg m 204 L n 300 mL
 o 328 kg
 2 b \$67.50 c \$57.60 d \$70 e 252 tonnes
 f 11.6 km g \$29.75 h \$12.60 i 64.5 L
 j \$7.68 k \$47.70 l \$8.76 m 14.4 L
 n 840 metres

EXTENSION IDEAS

Review the mind map the class created at the start of the topic, and have students add to it to show what they have learned.

You could also have students use percentage to explore classroom statistics, such as the percentage of students who walk to school, or have brown hair, or have surnames that begin with a vowel. Alternatively, students could interpret media and sports reports involving percentages.

1.1 Fractions of 100

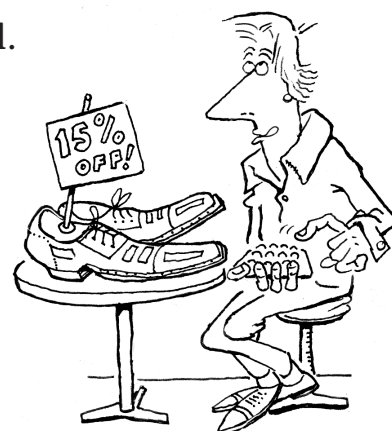
Many advertisements use percentages and the % symbol.
Your score in a test may be expressed as a percentage.

What does **percent** mean?

Percent means *out of each 100*.

A percentage is a fraction of 100.

80% means 80 out of 100 or $\frac{80}{100}$.



- 1 The teacher marked the test papers with scores out of 100. What **percentage** did these students score?

- a Nerida: 85 out of 100 = _____%
- b Sam: 60 out of 100 = _____%
- c Loi: 74 out of 100 = _____%

- 2 Write each piece of information as a fraction out of 100.
Then write it as a percentage.

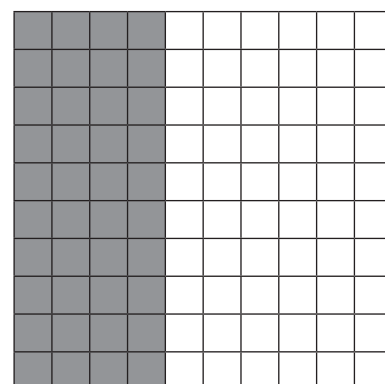
EXAMPLE

35 cents out of \$1 = $\frac{35}{100}$ of \$1 = 35 % of \$1

- a 80 cents out of \$1 = $\frac{\quad}{100}$ of \$1 = _____% of \$1
- b 95 cents out of \$1 = $\frac{\quad}{100}$ of \$1 = _____% of \$1
- c 30 cm out of 1 metre = $\frac{\quad}{100}$ of 1 m = _____% of 1 metre
- d 75 cm out of 1 metre = $\frac{\quad}{100}$ of 1 m = _____% of 1 metre

- 3 The grid at the right is divided into 100 squares.

- a How **many** squares are shaded? _____
- b What **fraction** of the grid is shaded? _____
- c What **percentage** of the grid is shaded? _____
- d What percentage of the grid is **not shaded**?

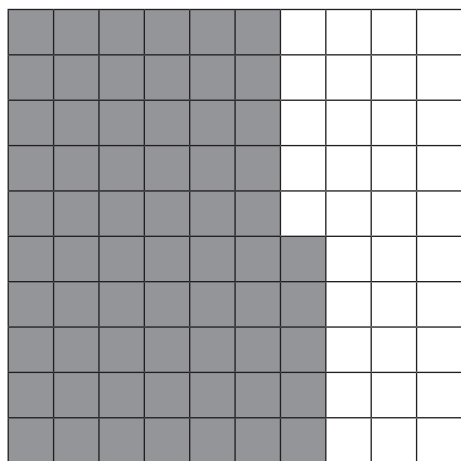


1.1 Fractions of 100

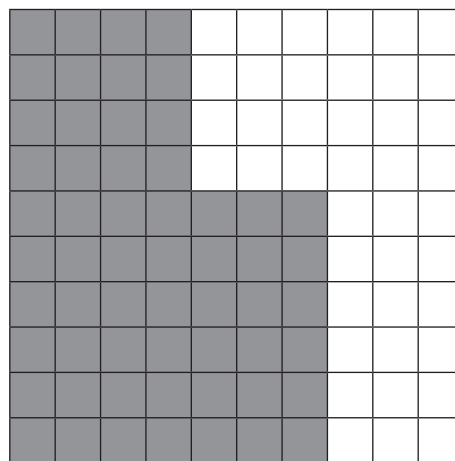
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4 What percentage of each grid is shaded?

a _____



b _____



5 Write these fractions as percentages.

a $\frac{25}{100} = 25\%$

b $\frac{15}{100} = \underline{\hspace{2cm}}$

c $\frac{60}{100} = \underline{\hspace{2cm}}$

d $\frac{30}{100} = \underline{\hspace{2cm}}$

e $\frac{90}{100} = \underline{\hspace{2cm}}$

f $\frac{75}{100} = \underline{\hspace{2cm}}$

g $\frac{42}{100} = \underline{\hspace{2cm}}$

h $\frac{86}{100} = \underline{\hspace{2cm}}$

i $\frac{5}{100} = \underline{\hspace{2cm}}$

6 Write these percentages as fractions with a denominator of 100.

a $40\% = \frac{40}{100}$

b $50\% = \underline{\hspace{2cm}}$

c $35\% = \underline{\hspace{2cm}}$

d $85\% = \underline{\hspace{2cm}}$

e $60\% = \underline{\hspace{2cm}}$

f $15\% = \underline{\hspace{2cm}}$

g $4\% = \underline{\hspace{2cm}}$

h $92\% = \underline{\hspace{2cm}}$

i $38\% = \underline{\hspace{2cm}}$

7 Each 100 grams of Fruity Flakes cereal contains about 35 grams of sugar.

What percentage is this? _____

8 In a survey, 80 boys out of 100 named football as their favourite sport.

What percentage is this? _____



1.2 Fractions and percentages

To convert a fraction to a percentage, multiply by 100%.

EXAMPLES

Express $\frac{3}{5}$ as a percentage.

$$\begin{aligned}\frac{3}{5} \times 100\% &= \frac{300}{5} \\ &= 60\%\end{aligned}$$

Convert $\frac{5}{8}$ to a percentage.

$$\begin{aligned}\frac{5}{8} \times 100\% &= \frac{500}{8} \\ &= 62\frac{1}{2}\%\end{aligned}$$

1 Convert these fractions to percentages.

a $\frac{4}{5}$ _____

b $\frac{3}{4}$ _____

c $\frac{2}{5}$ _____

d $\frac{3}{10}$ _____

e $\frac{7}{10}$ _____

f $\frac{9}{10}$ _____

g $\frac{11}{20}$ _____

h $\frac{9}{20}$ _____

i $\frac{17}{20}$ _____

j $\frac{1}{8}$ _____

k $\frac{3}{8}$ _____

l $\frac{1}{3}$ _____

m $\frac{2}{3}$ _____

n $\frac{5}{6}$ _____



1.3 Decimals and percentages

To convert a decimal to a percentage, multiply by 100%.

To do this, move the decimal point two places **to the right**.

EXAMPLES

Convert 0.3 to a percentage.

$$0.3 \times 100\% = 30\%$$

Change 0.345 to a percentage.

$$0.345 \times 100\% = 34.5\%$$

1 Convert these decimals to percentages.

a 0.14 _____ **b** 0.35 _____ **c** 0.72 _____

d 0.83 _____ **e** 0.06 _____ **f** 0.85 _____

g 0.4 _____ **h** 0.5 _____ **i** 0.3 _____

j 0.125 _____ **k** 0.625 _____ **l** 0.745 _____

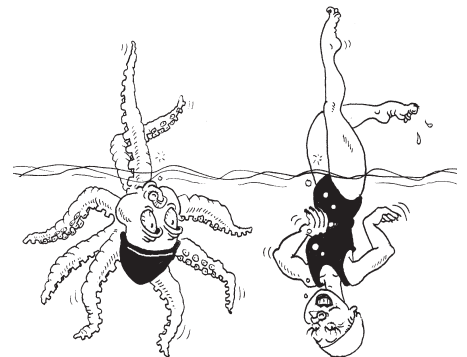
m 0.859 _____ **n** 0.106 _____ **o** 0.043 _____

p 0.48 _____ **q** 0.05 _____ **r** 0.115 _____

s 1.25 _____ **t** 1.65 _____

u 1.05 _____ **v** 1.5 _____

w 1.9 _____ **x** 2.5 _____



1.4 What percentage is that?

To express one quantity as a percentage of another quantity

- Step 1:** Check that both quantities are expressed in the **same units**.
- Step 2:** Write the first quantity as a **fraction** of the other.
- Step 3:** Change the fraction to a percentage by **multiplying** by 100%.

EXAMPLE

Express 75 cm as a percentage of 2 metres.

Step 1: Change 2 metres to 200 cm.

Step 2: Write $\frac{75}{200}$.

Step 3: $\frac{75}{200} \times 100\% = \frac{75}{2}$
 $= 37\frac{1}{2}\%$

- 1** Express \$15 as a percentage of \$20.

- 2** Express 28 cm as a percentage of 70 cm.

- 3** Express \$360 as a percentage of \$500.

- 4** Express 600 grams as a percentage of 2 kg.

- 5** Express 30 minutes as a percentage of 2 hours.

- 6** Express \$5.70 as a percentage of \$30.

Answer the following questions in your workbook.

- 7** In a survey, 50 girls out of 80 named netball as their favourite sport. What percentage is this?
- 8** A school has a total enrolment of 800 students. Yesterday 96 students were absent. What percentage of the students were absent?
- 9** A student received 21 marks in a maths test out of a total of 25 marks. What is her percentage mark?



1.5 Fractions, decimals and percentages

To convert a fraction or decimal to a percentage, **multiply** by 100%.

1 Convert these fractions to percentages.

a $\frac{27}{100} =$ _____ **b** $\frac{93}{100} =$ _____ **c** $\frac{11}{100} =$ _____

d $\frac{7}{10} =$ _____ **e** $\frac{2}{5} =$ _____ **f** $\frac{3}{4} =$ _____

g $\frac{1}{8} =$ _____ **h** $\frac{7}{8} =$ _____ **i** $\frac{2}{3} =$ _____

2 Convert these decimals to percentages.

a $0.62 =$ _____ **b** $0.37 =$ _____ **c** $0.25 =$ _____

d $0.8 =$ _____ **e** $0.5 =$ _____ **f** $0.2 =$ _____

g $0.04 =$ _____ **h** $0.125 =$ _____ **i** $0.875 =$ _____

To convert a percentage to a decimal, **divide** by 100.

To do this, move the decimal point two places **to the left**.

EXAMPLES

Convert 63% to a decimal.

$$63 \div 100 = 0.63$$

Change 37.5% to a decimal.

$$37.5 \div 100 = 0.375$$

3 Convert these percentages to decimals.

a $59\% =$ _____ **b** $75\% =$ _____ **c** $10\% =$ _____

d $35\% =$ _____ **e** $20\% =$ _____ **f** $50\% =$ _____

g $63\% =$ _____ **h** $80\% =$ _____ **i** $5\% =$ _____

j $12.5\% =$ _____ **k** $67.5\% =$ _____ **l** $9.5\% =$ _____

m $115\% =$ _____ **n** $125\% =$ _____ **o** $130\% =$ _____

1.5 Fractions, decimals and percentages

CONTINUED

To convert a percentage to a fraction, divide by 100 or multiply by $\frac{1}{100}$.

EXAMPLES

Convert 54% to a fraction.

$$\begin{aligned} 54\% &= 54 \times \frac{1}{100} \\ &= \frac{54}{100} \\ &= \frac{27}{50} \end{aligned}$$

Write $33\frac{1}{3}\%$ as a fraction.

$$\begin{aligned} 33\frac{1}{3}\% &= 33\frac{1}{3} \div 100 \\ &= \frac{100}{3} \times \frac{1}{100} \\ &= \frac{100}{300} \text{ or } \frac{1}{3} \end{aligned}$$



4 Convert these percentages to fractions. Simplify each answer where possible.

a $10\% = \frac{10}{100}$
 $= \frac{1}{10}$

b $25\% =$ _____

c $20\% =$ _____

d $40\% =$ _____

e $90\% =$ _____

f $75\% =$ _____

g $36\% =$ _____

h $15\% =$ _____

i $85\% =$ _____

j $5\% =$ _____

k $76\% =$ _____

l $92\% =$ _____

m $12\frac{1}{2}\% = 12\frac{1}{2} \div 100$
 $= \frac{25}{2} \times \frac{1}{100}$
 $= \frac{1}{8}$

n $37\frac{1}{2}\% =$ _____

o $66\frac{2}{3}\% =$ _____

p $120\% = \frac{120}{100}$
 $= \frac{6}{5} \text{ or } 1\frac{1}{5}$

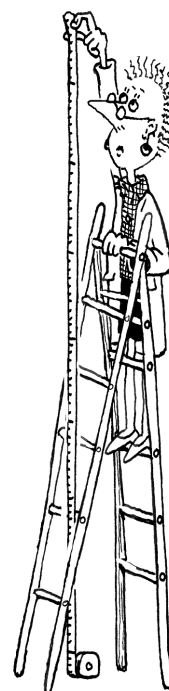
q $150\% =$ _____

r $175\% =$ _____

1.6 Equivalent fractions, decimals and percentages

Complete this table of equivalent fractions, decimals and percentages.

FRACTION	DECIMAL	PERCENTAGE
$\frac{1}{2}$	0.5	
$\frac{1}{3}$	$0.\dot{3}$	$33\frac{1}{3}\%$ or $33.\dot{3}\%$
$\frac{2}{3}$		$66\frac{2}{3}\%$ or $66.\dot{6}\%$
$\frac{1}{4}$		
$\frac{3}{4}$		
	0.2	20%
$\frac{2}{5}$		
		60%
	0.8	
$\frac{1}{8}$		
	0.375	
		$62\frac{1}{2}\%$ or 62.5%
$\frac{7}{8}$		
		10%
$\frac{3}{10}$		
	0.7	
		90%



1.7 Calculating a percentage A

The example shows one method for calculating a percentage of a quantity. Try doing this without a calculator.

The first step is to change the percentage to a fraction. Then multiply by the quantity.

EXAMPLE

Find 50% of \$34.

$$50\% = \frac{50}{100} = \frac{1}{2}$$

$$50\% \text{ of } \$34 = \frac{1}{2} \times 34 \text{ (or } 34 \div 2) \\ = \$17$$

1 Find 50% of these quantities.

a \$60

b \$72

c 150 km

2 Find 25% of these quantities.

a \$80

b \$56

c 176 kg

3 Find 10% of these quantities.

a \$120

b \$65

c 70 tonnes

4 Find 20% of these quantities.

a \$200

b \$95

c 55 metres

d 130 kg



1.8 Calculating a percentage B

A fashion store is having a sale.
All marked prices will be reduced by 20%.
How much will be taken off the price
of a dress marked at \$280?



To calculate a percentage of a quantity,
we use this rule:

$$\frac{\text{Percentage}}{100} \times \text{Quantity}$$

or $\text{Percentage} \div 100 \times \text{Quantity}$

EXAMPLE

Calculate 20% of \$280.

$$\begin{aligned} 20\% \text{ of } \$280 &= \frac{20}{100} \times 280 \\ &= \$56 \end{aligned}$$

1 Calculate.

a 30% of \$80

b 60% of \$120

c 20% of \$350

d 70% of \$300

e 40% of \$160

f 5% of \$200

g 25% of \$380

h 15% of 300 metres

i 35% of 400 kg

j 70% of 1 200 km

k 80% of 450 grams

l 90% of 3 200 kg

m 60% of 340 litres

n 40% of 750 mL

o 80% of 410 kg

1.8 Calculating a percentage B

CONTINUED

2 Use a calculator for these calculations.

Write out the calculation and your answer, as shown for part a.

a 60% of \$390

$$= 60 \div 100 \times 390$$

$$= \$234$$

b 15% of \$450

c 32% of \$180

d 28% of \$250

e 90% of 280 tonnes

f 4% of 290 km

g 35% of \$85

h 18% of \$70

i 43% of 150 litres

j 60% of \$12.80

k 53% of \$90

l 24% of \$36.50

m 120% of 12 litres

n 105% of 800 metres

Teacher's notes

OBJECTIVES

- Increase a quantity by a given percentage.
- Decrease a quantity by a given percentage.
- Solve problems relating to discounts.
- Solve problems relating to percentage composition.
- Calculate the percentage profit or loss when an item is sold.

PRIOR KNOWLEDGE

Students should be able to perform basic calculations with percentages. They should also be able to calculate a percentage of a quantity and express one quantity as a percentage of another.

BACKGROUND

Most of these activity sheets have been developed for students to complete without using a calculator. It may be appropriate for students to complete some activity sheets using pen and paper methods and then check their answers using a calculator. This will reinforce both pen and paper and calculator skills.

When calculators are being used, it is important that students write at least one line of working to show the calculation that they have performed.

Some students may need assistance with the subtleties of the English language when solving problems involving percentages. For example, *10% of \$40* is **not** the same as *10% off \$40*.

STARTER ACTIVITY

Hold a brief mental maths quiz, challenging students to give equivalent fractions for common percentage and vice versa; for example, 10%, 50%, 25%. Then lead a terminology discussion to help students think about whether an answer is going to be more or less than the given quantity. For example, '\$100 is increased by 10%. Will the answer be more or less than \$100?' Review terms such as *increase by*, *decrease by*, *discount*, *off*, *of*, *marked up by*, *marked down by*.

ACTIVITY SHEETS

2.1 Percentage increase

This activity sheet develops students' skills in increasing a quantity by a given percentage. Two methods are shown in the example. You should direct each student to **one** of these methods, depending on the student's level of competency.

Answers

- 1 a \$72 b \$180 c \$276 d 900 mL e 96 kg
f 5.76 m
- 2 a \$55 b \$224 c \$54.60 d \$210
e \$793.50 f \$31.35 g 112 km h 63 tonnes
i 3.68 kg j \$135
- 3 \$149.50
- 4 \$100.80
- 5 \$762.20

2.2 Percentage decrease

This activity sheet develops students' skills in decreasing a quantity by a given percentage. Two methods are shown in the example. You should direct each student to **one** of these methods, depending on the student's level of competency.

Answers

- 1 a \$72 b \$117 c \$58.50 d 180 km
e 405 L f 8.82 kg
- 2 a \$176 b \$102 c \$255 d \$942 e \$516
f \$17.10 g 66 m h 128 kg i 59.5 L
j 175 km
- 3 16 928
- 4 \$1 295
- 5 \$626.25

2.3 Discounts

In this activity sheet, students will apply their skills in decreasing by a percentage to problems involving discounts.

Answers

- 1 a \$16 b \$10 c \$60 d \$36 e \$3 f \$42
- 2 a \$54, \$486 b \$23.50, \$211.50
c \$379, \$1 516 d \$64.50, \$1 225.50
e \$54.90, \$494.10 f \$129, \$731
- 3 \$36.50
- 4 a \$56 b \$224
- 5 \$117.30
- 6 \$1 775.55

Teacher's notes

CONTINUED

2.4 Percentage composition

In this activity sheet, students will look at a number of examples where percentages are used to express the components of a mixture. Students will need to calculate percentages of quantities and express one quantity as a percentage of another. Most students will need to use a calculator to complete this activity sheet.

Answers

- 1 495 g
 2 87 g
 3 15 kg
 4 a 264 g b 28 g
 5 4.3%
 6 11.4%
 7 50% copper, 25% zinc, 25% nickel
 8 a 45.1% b 3.0%

2.5 Percentage profit

In this activity sheet, students look at the profit when an item is sold. They will calculate the profit (by subtraction) and then calculate the percentage profit, based on the cost price. Students will need to use a calculator to complete this activity sheet.

Answers

- 1 a \$15 b 25%
 2 a \$23 b 50%
 3 a \$36 b 20%
 4 a \$95 b 25%
 5 a \$5.60 b 40%
 6 a \$6.30 b 35%
 7 a 29.7% b 51.7% c 32.6% d 27.1%
 e 15.4% f 22.4%
 8 35%
 9 45.8%
 10 36%

2.6 Percentage loss

In this activity sheet, students look at a loss when an item is sold. They will calculate the loss (by subtraction) and then calculate the percentage loss, based on the cost price. Students will need to use a calculator to complete this activity sheet.

Answers

- 1 a \$15 b 30%
 2 a \$15 b 25%
 3 a \$24 b 20%
 4 a \$70 b 35%
 5 a \$116 b 40%
 6 a \$13.50 b 45%
 7 a 23.1% b 11.6% c 35.8% d 19.7%
 e 36.7% f 44.4%
 8 45%
 9 35%
 10 34.4%

EXTENSION IDEAS

Provide packaging for three similar products, such as packets of three different brands of potato chips. Have the students convert the main ingredients to percentages in order to compare the three different brands. Encourage them to compose general statements comparing each main ingredient.

2.1 Percentage increase

The price of a designer T-shirt was \$85.

The shop increases the price by 20%.

To calculate the new price, we increase \$85 by 20%.

When a quantity is increased by a given percentage, it is made larger. The percentage amount is **added** to the original quantity.



EXAMPLE

Increase \$85 by 20%.

Method 1:

$$\begin{aligned} 20\% \text{ of } \$85 &= \frac{20}{100} \times 85 \\ &= \$17 \end{aligned}$$

$$\begin{aligned} \text{The increased amount is } \$85 + \$17 \\ &= \$102 \end{aligned}$$

Method 2:

The new amount is $100\% + 20\%$
 $= 120\%$ of the original

$$\begin{aligned} 120\% \text{ of } \$85 &= \frac{120}{100} \times 85 \\ &= \$102 \end{aligned}$$

1 Increase each amount by 20%.

a \$60

b \$150

c \$230

d 750 mL

e 80 kg

f 4.8 metres

2 Complete these percentage increase calculations.

a Increase \$50 by 10%.

b Increase \$200 by 12%.

2.1 Percentage increase

CONTINUED

c Increase \$42 by 30%.

d Increase \$168 by 25%.

e Increase \$690 by 15%.

f Increase \$28.50 by 10%.

g Increase 80 km by 40%.

h Increase 60 tonnes by 5%.

i Increase 3.2 kg by 15%.

j Increase \$120 by 12.5%.

Answer the following questions in your workbook.

- 3** A pair of sports shoes is priced at \$130.
What is the new price if the shop increases all prices by 15%?
- 4** A satchel cost \$72 to make.
Calculate the selling price if the cost is marked up by 40%.
- 5** Jamal's current weekly pay is \$740.
His pay is increased by 3%.
What is his new weekly pay?



2.2 Percentage decrease

A skateboard is priced at \$230.

During a sale, the price is reduced by 10%.

To calculate the sale price, we **decrease \$230 by 10%**.

When a quantity is decreased by a given percentage, it is made smaller. The percentage amount is **subtracted** from the original quantity.



... EXAMPLE

Decrease \$230 by 10%.

Method 1:

$$10\% \text{ of } \$230 = \frac{10}{100} \times 230$$

$$= \$23$$

$$\text{The decreased amount is } \$230 - \$23$$

$$= \$207$$

Method 2:

The new amount is $100\% - 10\%$
 $= 90\%$ of the original

$$90\% \text{ of } \$230 = \frac{90}{100} \times 230$$

$$= \$207$$

1 Decrease each amount by 10%.

a \$80

b \$130

c \$65

d 200 km

e 450 litres

f 9.8 kg

2 Complete these percentage decrease calculations.

a Decrease \$200 by 12%.

b Decrease \$120 by 15%.

2.2 Percentage decrease

CONTINUED

c Decrease \$340 by 25%.

d Decrease \$1 570 by 40%.

e Decrease \$645 by 20%.

f Decrease \$18 by 5%.

g Decrease 120 metres by 45%.

h Decrease 160 kg by 20%.

i Decrease 70 litres by 15%.

j Decrease 200 km by 12.5%.

Answer the following questions in your workbook.

- 3** The population of a town was 18 400. It has decreased by 8%.
What is the current population of this town?
- 4** A white-water rafting weekend is priced at \$1 850.
During a sale, the price is reduced by 30%.
What is the sale price?
- 5** Katie's weekly wage is \$835. This is decreased
by 25% when income tax is deducted.
How much remains?



2.3 Discounts

Mario bought a computer priced at \$1 490.
He received a discount of 5% for paying cash.

The normal price of an item is sometimes reduced.
The amount by which the price is reduced is called the **discount**. This is often given as a **percentage**.



EXAMPLE

Marco received a 5% discount
on a computer priced at \$1 490.

How much did he save?
(What is the amount
of the discount?)

$$\begin{aligned}\text{amount saved} &= \frac{\text{percentage}}{100} \times \text{price} \\ &= \frac{5}{100} \times 1\,490 \\ &= \$74.50\end{aligned}$$

1 Calculate the amount of these discounts.

a 10% on \$160

b 20% on \$50

c 25% on \$240

d 20% on \$180

e 5% on \$60

f 15% on \$280

2 Calculate the **value** of each discount and the **discounted price** of each item.
(Calculate the amount of the discount. Then **subtract** this from the original price.)

a A washing machine priced at \$540, discount 10%

b A microwave oven priced at \$235, discount 10%

2.3 Discounts

CONTINUED

- c** A television priced at \$1 895,
discount 20%

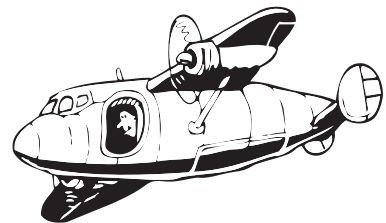
- d** A dishwasher priced at \$1 290,
discount 5%

- e** A car GPS priced at \$549,
discount 10%

- f** A set of car tyres priced at \$860,
discount 15%

Answer the following questions in your workbook.

- 3** A skydive is normally priced at \$365.
Perrine gets a 10% discount on this price.
How much does she save?
- 4** At a sale, all prices are discounted by 20%.
Charlotte buys a watch priced at \$280.
- a** How much does she save?
- b** How much does Charlotte pay for the watch?
- 5** Richard bought a sports bag priced at \$138.
He was given a discount of 15% on this price.
How much did Richard pay for the sports bag?
- 6** Thoi bought a refrigerator priced at \$1 869.
She received a discount of 5% for paying cash.
How much did Thoi pay?



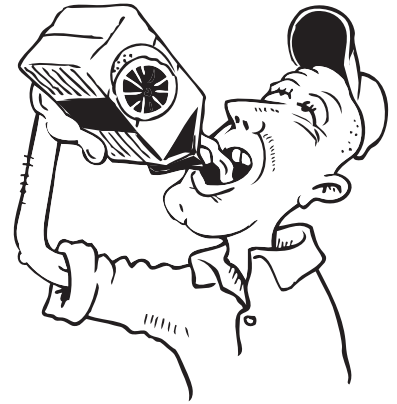
2.4 Percentage composition

Percentages are often used to indicate the ingredients or the nutrition information of a product.

EXAMPLE

A 2 litre carton of fruit drink states that it contains 35% orange juice. How many millilitres of orange juice does this carton contain?

$$\begin{aligned} 35\% \text{ of } 2 \text{ litres} &= \frac{35}{100} \times 2\,000 \text{ mL} & (2 \text{ L} = 2\,000 \text{ mL}) \\ &= 700 \text{ mL} \end{aligned}$$



- 1** An 825 gram can of fruit salad contains 60% fruit.

What is the mass of the fruit in this can?

- 2** A packet of dog biscuits contains 58% protein.

How much protein is in a 150 gram packet of biscuits?

- 3** Approximately 75% of the mass of potatoes is water.

What is the mass of water in 20 kg of potatoes?

- 4** Part of the ingredients list on a 400 gram jar of pasta sauce is shown in this table.

Chopped tomatoes	66%
Tomato concentrate	10%
Onions	7%
Water	3%

- a** What is the mass of the chopped tomatoes in this jar?

- b** What is the mass of the onions in this jar?

2.4 Percentage composition

CONTINUED

EXAMPLE

The following information is printed on a packet of breakfast cereal.

Every 30 grams contains:	
Protein	3.7 g
Fat	0.4 g
Carbohydrate	20.1 g
Dietary fibre	3.3 g

What percentage of the contents is protein?

$$\frac{3.7}{30} \times 100$$

$$= 12.3\% \text{ (rounded to 1 decimal place)}$$

Answer the following questions correct to 1 decimal place.

- 5** A 200 gram packet of rice crackers contains 8.6 grams of fat.

What percentage of the contents is fat?

- 6** An 825 gram can of fruit contains 94 grams of sugars.

What percentage of the contents is sugars?

- 7** A nickel alloy contains 2 parts copper, 1 part zinc and 1 part nickel.

Express the composition of this alloy using percentages.

- 8** The information below is printed on a wrapper for sliced bread.

Each 72.5 gram serving contains:	
Protein	5.8 g
Fat	1.8 g
Carbohydrate	32.7 g
Dietary fibre	2.2 g

- a** What percentage of each serving is carbohydrate?

- b** What percentage of each serving is dietary fibre?



2.5 Percentage profit

EXAMPLE

A jeweller purchases a ring for \$185. He sells it for \$259.

What is the percentage profit, based on the cost price?

- The **cost price** is how much a store pays for an item.
- The **selling price** is how much is received when the item is sold.
- If I sell an item for **more** than I paid for it, I make a **profit**.

profit = selling price – cost price

percentage profit (based on the cost price)

$$= \frac{\text{profit}}{\text{cost price}} \times 100\%$$

$$\text{cost price} = \$185$$

$$\text{selling price} = \$259$$

$$\begin{aligned} \text{profit} &= \$259 - \$185 \\ &= \$74 \end{aligned}$$

$$\text{percentage profit}$$

$$= \frac{74}{185} \times 100$$

$$= 40\%$$

For each of the following items, calculate:

- the amount of the profit
- the percentage profit, based on the cost price.

- cost price = \$60
selling price = \$75

- cost price = \$46
selling price = \$69

- cost price = \$180
selling price = \$216

- cost price = \$380
selling price = \$475



2.5 Percentage profit

CONTINUED

- 5** cost price = \$14
selling price = \$19.60

- 6** cost price = \$18
selling price = \$24.30

Answer the following questions in your workbook.

- 7** For the following items, calculate the percentage profit, based on the cost price. Give each answer correct to 1 decimal place.

- a** cost price = \$74
selling price = \$96



- b** cost price = \$89
selling price = \$135



- c** cost price = \$215
selling price = \$285



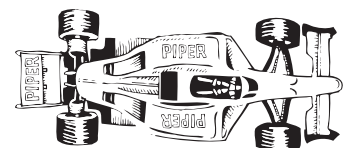
- d** cost price = \$55
selling price = \$69.90



- e** cost price = \$3.90
selling price = \$4.50



- f** cost price = \$12.25
selling price = \$15



- 8** A shop buys a rug for \$540 and sells it for \$729.
Calculate the percentage profit, based on the cost price.
- 9** A furniture store purchased a bookcase for \$72. It was sold for \$105.
Calculate the percentage profit, based on the cost price.
Give your answer correct to 1 decimal place.
- 10** A used car yard sold a car for \$14 990. They had paid \$11 000 for this car.
Calculate the percentage profit, based on the cost price.
Give your answer to the nearest whole number.

2.6 Percentage loss

EXAMPLE

Lucile bought a bicycle for \$360. One year later, she sold it for \$270.

What is the loss as a percentage of the cost price?

- The **cost price** is how much you paid for an item.
- The **selling price** is how much you receive when the item is sold.
- If I sell an item for **less** than I paid for it, I make a **loss**.

loss = cost price – selling price

percentage loss (based on the cost price)

$$= \frac{\text{loss}}{\text{cost price}} \times 100\%$$

cost price = \$360

selling price = \$270

loss = \$360 – \$270
= \$90

percentage loss

$= \frac{90}{360} \times 100$
= 25%

The following items were sold at a loss. Calculate:

- the amount of the loss
- the percentage loss, based on the cost price.

- 1** cost price = \$50
selling price = \$35

- 2** cost price = \$60
selling price = \$45

- 3** cost price = \$120
selling price = \$96

- 4** cost price = \$200
selling price = \$130

2.6 Percentage loss

CONTINUED

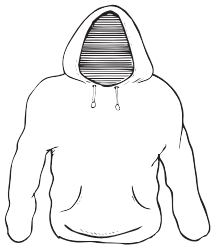
- 5** cost price = \$290
selling price = \$174

- 6** cost price = \$30
selling price = \$16.50

Answer the following questions in your workbook.

- 7** The following items were sold at a loss.
For each item, calculate the percentage loss, based on the cost price.
Give each answer correct to 1 decimal place.

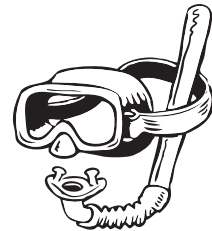
- a** cost price = \$65
selling price = \$50



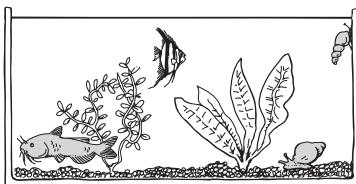
- b** cost price = \$95
selling price = \$84



- c** cost price = \$179
selling price = \$115



- d** cost price = \$380
selling price = \$305



- e** cost price = \$15
selling price = \$9.50



- f** cost price = \$18.90
selling price = \$10.50



- 8** Brian bought a bicycle for \$540. He sold it one year later for \$297.
Calculate the percentage loss, based on the cost price.
- 9** Kate purchased a set of golf clubs for \$890. She later sold them for \$580.
Calculate the percentage loss, based on the cost price.
Give your answer to the nearest whole number.
- 10** Melina purchased a computer for \$1 295. She later sold it for \$850.
Calculate the percentage loss, based on the cost price.
Give your answer correct to 1 decimal place.

Teacher's notes

OBJECTIVES

- Understand the concept of a ratio.
- Complete equivalent ratios.
- Simplify a ratio.
- Use equivalent ratios to solve problems.
- Divide a quantity in a given ratio.
- Solve problems involving ratio.

PRIOR KNOWLEDGE

The following activity sheets have been developed for students to complete without using calculators. Students should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or pen and paper methods.

BACKGROUND

Many real-life situations use ratios. Examples include scales on maps; scales in model making; mixes of concrete, alloys and fuels; and gear ratios in machines.

It may be appropriate for students to complete some activity sheets using pen and paper methods and then check their answers using a calculator. This will reinforce both pen and paper and calculator skills.

STARTER ACTIVITY

Provide approximately 100 physical objects such as counters, pasta pieces or Smarties. Start with a small number, such as 15. Challenge two students to share the objects so that one student has twice as many as the other student. Have the students brainstorm different techniques for achieving this. Repeat with different pairs of students and different sharing proportions. Then ask students to think about how they could work out the shares of large amounts – such as 1 000 – without counting out each object.

ACTIVITY SHEETS

3.1 What is a ratio?

This activity sheet introduces a ratio to compare two quantities of the same kind; for example, two lengths or two amounts of money. It is not intended that students simplify their answers in this activity sheet. Emphasise to students that correct order of the ratio is essential. For example, in the first activity, the ratio of shaded squares to unshaded squares is 3:5, **not** 5:3

Answers

- 1 a 3 b 5
 2 a 5:3 b 5:4 c 5:7 d 1:3 e 7:3 f 3:5
 3 a 4:5 b 50:70 c 10:5 d 20:15 e 5:12
 f 6:5 g 12:18 h 130:145 i 15:25 j 12:7
 4 a AB = 6 cm, CD = 8 cm b 6:8
 5 a 3:5 b 5:2 c 1:20 d 1:3 e 2:1
 6 a 50:100 b 750:1 000 c 30:100 d 20:60
 e 200:1 000 f 650:2 000 g 75:400
 h 4 000:920 i 2 000:960 j 30:15

3.2 Equivalent ratios

This activity sheet develops students' understanding of ratio to include multiplying and dividing to find equivalent ratios.

Answers

- 1 a 6:2 b 9:3 c 3:1
 2 a 21 b 16 c 12 d 40 e 10 f 9 g 4
 h 12 i 20
 3 a 6 b 4 c 4 d 1 e 15 f 4 g 3
 h 5 i 2

3.3 Using equivalent ratios

In this activity sheet, students use equivalent ratios to solve simple problems. Encourage students to write working that shows two equivalent ratios, rather than just an answer.

Answers

- 1 20 km
 2 20
 3 25
 4 \$16
 5 120

Teacher's notes

CONTINUED

3.4 Simplifying ratios

This activity sheet develops students' skills in simplifying a given ratio. Some students may need to be reminded about common factors before they start their work.

Answers

1 a 1:4 b 2:3 c 3:4 d 1:2 e 3:1
f 2:1 g 3:5 h 4:5 i 1:3 j 1:3
k 5:4 l 2:1 m 1:3 n 1:3 o 9:13
2 a 2:3 b 3:4 c 2:3 d 3:5 e 3:4 f 5:7
g 7:9 h 2:5
3 a 4:5 b 9:10 c 4:1 d 5:12 e 16:9
f 8:3 g 4:3 h 3:8 i 2:1 j 4:15
4 a 3:5 b 1:3 c 5:6 d 10:7 e 5:7
f 3:2 g 3:2 h 2:3 i 1:3

3.5 Increase in a ratio

This activity sheet is an application of equivalent ratios. The use of proportions is an alternate method for solving these problems. Encourage students to write working that shows two equivalent ratios, rather than just an answer.

Answers

1 a \$32 b \$120 c 24 kg d \$72 e 500 mL
f \$150
2 a \$420 b 90 mL c \$6.30 d 540

3.6 Decrease in a ratio

This activity sheet is another application of equivalent ratios. Encourage students to write working that shows two equivalent ratios, rather than just an answer.

Answers

1 a \$7 b 10 cm c \$12 d 18 kg e \$150
f 48 kg
2 a \$66 b 72 metres c \$6.30 d \$45

3.7 Dividing in a ratio

This activity sheet looks at an important ratio skill – dividing a given quantity according to a given ratio. The example shows one method that can be applied. Some students may need to be reminded how to calculate a fraction of a quantity. It is important that students write their answers in the correct order, as determined by the ratio.

Answers

1 a 5 b 4 c 7 d 8 e 9 f 11
2 a \$16, \$8 b \$8, \$24 c \$20, \$30
d \$18, \$30 e \$24, \$18 f \$48, \$32
g \$20, \$35 h \$125, \$50 i 24 cm, 96 cm
j 200 mL, 300 mL k 70 m, 80 m
l 75 kg, 105 kg m 150 km, 100 km
n 320 mm, 400 mm
3 600 g, 400 g
4 25 min, 35 min
5 \$16
6 \$120
7 \$21
8 119
9 100 square metres
10 \$12 500
11 \$1 000

3.8 Dividing into 3 parts

This activity sheet is provided as an extension activity. Students encounter ratios that involve three parts rather than two.

Answers

1 \$8, \$16, \$24
2 \$8, \$24, \$40
3 \$48, \$36, \$12
4 \$40, \$60, \$100
5 \$50, \$100, \$75
6 \$80, \$80, \$120
7 \$165, \$55, \$220
8 \$145, \$232, \$203
9 \$110
10 \$175

EXTENSION IDEAS

Have students solve a variety of real-life problems involving ratios; for example, mixes for fuels or concrete, scales on maps and gear ratios on machines.

Students could investigate Golden Rectangles. The ratio of the dimensions of a Golden Rectangle was known to the Ancient Greeks.

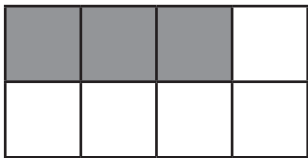
Length : width is approximately 1.62 : 1 (or $\text{length} \div \text{width} = 1.62$ correct to 2 decimal places).

Many windows are golden rectangles. Students could check the dimensions of the classroom windows.

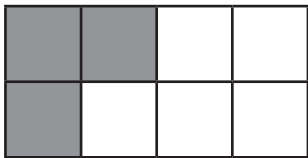
3.1 What is a ratio?

A **ratio** compares two or more quantities of the same kind. The order is important and the quantities must be expressed in the same units.

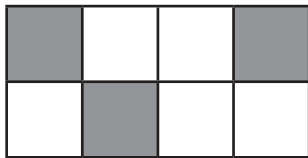
- 1 Look at the diagrams below.
 - a How many squares are **shaded** in each diagram?
 - b How many squares are **not shaded** in each diagram?



Shaded: 3
Not shaded: 5



Shaded:
Not shaded:



Shaded:
Not shaded:

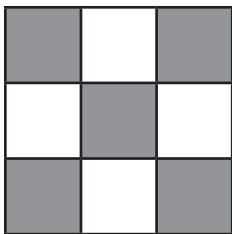
In each case, 3 squares are shaded and 5 squares are not shaded. The **ratio** of shaded squares to unshaded squares is **3 to 5**. A ratio can be written in words, **3 to 5**, or using a colon, **3:5**.

- 2 Write the ratio of shaded parts to unshaded parts for each diagram. Use a colon (:).

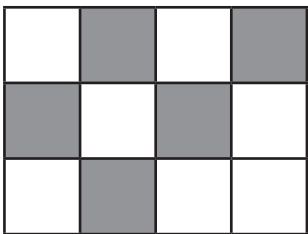
a 5:3



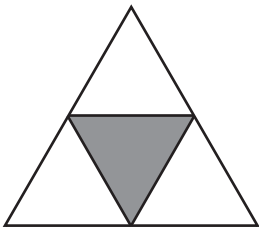
b



c



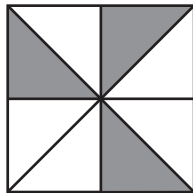
d



e



f



3.1 What is a ratio?

CONTINUED

- 3** Write a ratio to describe these comparisons.
Write the numbers only. A ratio has **no units**.

- | | |
|---------------------------------------|-------------------------------------|
| a 4 boys to 5 girls <u>4:5</u> | b 50 cents to 70 cents _____ |
| c 10 years to 5 years _____ | d \$20 to \$15 _____ |
| e 5 metres to 12 metres _____ | f 6 women to 5 men _____ |
| g 12 points to 18 points _____ | h 130 cm to 145 cm _____ |
| i 15 kg to 25 kg _____ | j 12 games to 7 games _____ |

- 4** The diagrams below show two intervals – AB and CD.



- a** Measure the length of each interval to the nearest centimetre.

AB _____ CD _____

- b** Write the ratio of the lengths AB:CD. _____

- 5** Write the ratio suggested by each statement.

- a** There were 3 boys present for every 5 girls. _____
- b** There were 5 children for every 2 adults. _____
- c** For each teacher, there were 20 students. _____
- d** For each person standing, there were 3 people seated. _____
- e** This book costs twice as much as that book. _____



- 6** Change the quantities to the same units. Then write each pair as a ratio.

- | | |
|---|-------------------------------------|
| a 50 cm to 1 metre = <u>50:100</u> | b 750 grams to 1 kg _____ |
| c 30 cents to \$1 _____ | d 20 minutes to 1 hour _____ |
| e 200 kg to 1 tonne _____ | f 650 mL to 2 litres _____ |
| g 75 cents to \$4 _____ | h 4 km to 920 metres _____ |
| i 2 kg to 960 grams _____ | j 3 cm to 15 mm _____ |

3.2 Equivalent ratios

The ratio is 9 : 3.



That's the same as a ratio of 3 : 1!

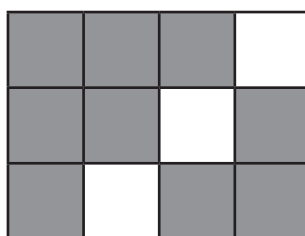
Annaka said the ratio was 9 : 3.
Ryan said this was the same as 3 : 1.
Is Ryan correct?

1 For each diagram, write the ratio of shaded squares to unshaded squares.

a _____



b _____



c _____



In the diagrams above, there are 3 shaded squares to 1 unshaded square in each row. There are 3 shaded squares for every unshaded square in each diagram. Shaded squares and unshaded squares are in the ratio 3 : 1.

The ratios 6 : 2, 9 : 3 and 3 : 1 are **equivalent**. Equivalent ratios are formed if you multiply or divide **both** terms by the **same number**.

EXAMPLES

$$3 : 1 = 3 \times 4 : 1 \times 4 \\ = 12 : 4$$

$$8 : 12 = 8 \div 4 : 12 \div 4 \\ = 2 : 3$$

2 Use **multiplication** to complete these equivalent ratios.

a $2 : 7 = 6 : \underline{\hspace{1cm}}$

b $3 : 8 = 6 : \underline{\hspace{1cm}}$

c $3 : 4 = 9 : \underline{\hspace{1cm}}$

d $3 : 10 = 12 : \underline{\hspace{1cm}}$

e $5 : 2 = 25 : \underline{\hspace{1cm}}$

f $1 : 3 = 3 : \underline{\hspace{1cm}}$

g $2 : 5 = \underline{\hspace{1cm}} : 10$

h $4 : 7 = \underline{\hspace{1cm}} : 21$

i $5 : 3 = \underline{\hspace{1cm}} : 12$

3 Use **division** to complete these equivalent ratios.

a $10 : 12 = 5 : \underline{\hspace{1cm}}$

b $15 : 20 = 3 : \underline{\hspace{1cm}}$

c $12 : 16 = 3 : \underline{\hspace{1cm}}$

d $24 : 6 = 4 : \underline{\hspace{1cm}}$

e $8 : 30 = 4 : \underline{\hspace{1cm}}$

f $18 : 24 = 3 : \underline{\hspace{1cm}}$

g $12 : 4 = \underline{\hspace{1cm}} : 1$

h $45 : 27 = \underline{\hspace{1cm}} : 3$

i $22 : 55 = \underline{\hspace{1cm}} : 5$

3.3 Using equivalent ratios

EXAMPLE

Two tickets to a concert cost \$90.
What would be the cost of three tickets to this concert?

Ratio of tickets = ratio of cost

$$2 : 3 = 90 : \underline{\hspace{1cm}}$$

$$90 = 2 \times 45 \quad (90 \div 2 = 45)$$

$$\begin{aligned} \text{So } 2 : 3 &= 2 \times 45 : 3 \times 45 \\ &= 90 : 135 \end{aligned}$$

The cost is \$135.

The problems on this activity sheet can all be solved using equivalent ratios.

- 1 Keiko walked 12 km in 3 hours.
At the same pace, how far would she walk in 5 hours?

- 2 The ratio of boys to girls at a party was 3 : 4. There were 15 boys. How many girls were at the party?

- 3 The ratio of Carlo's age to Christian's age is 5 : 4.
Christian is 20 years old.
How old is Carlo?

- 4 The ratio of the profit to the selling price of an item was 2 : 5.
The selling price was \$40.
What was the profit?

- 5 Neelam counted the cars travelling north and south along a freeway. The ratio for travelling north to travelling south was 9 : 5. Neelam counted 216 cars travelling north. How many cars did she count travelling south?

3.4 Simplifying ratios

We **simplify** a ratio when we divide both terms by any **common factors**.
A common factor will divide both terms without a remainder.

EXAMPLES

Simplify 24 : 30.

24 : 30 (divide both terms by 6)
= 4 : 5

Simplify 150 : 120.

150 : 120 (divide both terms by 10)
= 15 : 12 (divide both terms by 3)
= 5 : 4

1 Simplify each ratio.

- a** 2 : 8 = _____

d 5 : 10 = _____

g 9 : 15 = _____

j 15 : 45 = _____

m 16 : 48 = _____
- b** 6 : 9 = _____

e 12 : 4 = _____

h 12 : 15 = _____

k 30 : 24 = _____

n 80 : 240 = _____
- c** 6 : 8 = _____

f 18 : 9 = _____

i 8 : 24 = _____

l 36 : 18 = _____

o 270 : 390 = _____

2 Simplify each ratio. Remember, a ratio has no units.

- a** 10 cm : 15 cm = _____

c 12 m : 18 m = _____

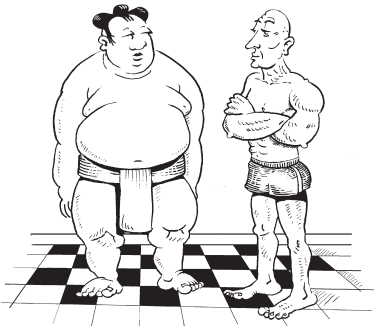
e \$24 : \$32 = _____

g 140 m : 180 m = _____

h 300 g : 750 g = _____
- b** \$15 : \$20 = _____

d 15 kg : 25 kg = _____

f 50 mm : 70 mm = _____



3.4 Simplifying ratios

CONTINUED

3 Change the quantities to the same units. Then simplify each ratio.

a 80 cm : 1 metre = 80 : 100

= 4 : 5

b 90 cm : 1 metre = _____

c 2 litres : 500 mL = _____

d 10 hours : 1 day = _____

e 8 cm : 45 mm = _____

f 2 kg : 750 g = _____

g 80 seconds : 1 minute = _____

h 75 cents : \$2 = _____

i 5 kg : 2 500 g = _____

j 800 mL : 3 litres = _____

If a ratio involves decimals, first **multiply** both terms by 10, 100 or 1 000 to remove the decimal points.

EXAMPLES

Simplify 1.5 : 1.25.

1.5 : 1.25 (multiply by 100)

= 150 : 125 (divide by 5)

= 30 : 25 (divide by 5 again)

= 6 : 5

Simplify \$2.50 : \$1.50.

\$2.50 : \$1.50 (remove units)

= 2.5 : 1.5 (multiply by 10)

= 25 : 15 (divide by 5)

= 5 : 3

Answer these questions in your workbook.

4 Simplify each ratio.

a 1.5 : 2.5

b 1.2 : 3.6

c 1.25 : 1.5

d 2.5 : 1.75

e \$2.50 : \$3.50

f \$2.40 : \$1.60

g \$4.20 : \$2.80

h 2.4 kg : 3.6 kg

i 1.25 km : 3.75 km

3.5 Increase in a ratio

Increasing a quantity in a given ratio is an application of equivalent ratios. Equivalent ratios are formed when you multiply both terms by the same number.

EXAMPLE

Increase 50 cm in the ratio 2 : 3.

This question can be rewritten as:

Complete the equivalent ratios $2 : 3 = 50 : \underline{\hspace{1cm}}$.

$$2 : 3 = 2 \times 25 : 3 \times 25$$
$$= 50 : 75$$

The answer is 75 cm.

($50 = 2 \times 25$, so we must multiply 3 by 25 as well)

1 Increase these quantities.

- a Increase \$16 in the ratio 1 : 2.

- b Increase \$80 in the ratio 2 : 3.

- c Increase 18 kg in the ratio 3 : 4.

- d Increase \$27 in the ratio 3 : 8.

- e Increase 200 mL in the ratio 2 : 5.

- f Increase \$120 in the ratio 4 : 5.

2 Answer the following questions in your workbook.

- a Increase \$240 in the ratio 4 : 7.
- b Increase 75 mL in the ratio 5 : 6.
- c Increase \$5.40 in the ratio 6 : 7.
- d Last year Katie scored a total of 216 runs at cricket. She has improved a lot this year. The total number of runs Katie scored increased in the ratio 2 : 5. How many runs did Katie score this year?

3.6 Decrease in a ratio

Decreasing a quantity in a given ratio is an application of equivalent ratios. Equivalent ratios are formed when you multiply both terms by the same number.

... **EXAMPLE**

Decrease \$20 in the ratio 4 : 3.

This question can be rewritten as:

Complete the equivalent ratios 4 : 3 = 20 : ____.

$4 : 3 = 4 \times 5 : 3 \times 5$
 $= 20 : 15$

The answer is \$15.

(20 = 4 × 5, so we must multiply 3 by 5 as well)

1 Decrease these quantities.

- a** Decrease \$14 in the ratio 2 : 1.

- b** Decrease 15 cm in the ratio 3 : 2.

- c** Decrease \$30 in the ratio 5 : 2.

- d** Decrease 24 kg in the ratio 4 : 3.

- e** Decrease \$180 in the ratio 6 : 5.

- f** Decrease 56 kg in the ratio 7 : 6.

2 Answer the following questions in your workbook.

- a** Decrease \$176 in the ratio 8 : 3.
- b** Decrease 126 metres in the ratio 7 : 4.
- c** Decrease \$10.50 in the ratio 5 : 3.
- d** During a sale, the price of a pair of boardshorts decreased in the ratio 4 : 3. The original price was \$60. What was the price during the sale?

3.7 Dividing in a ratio

Franco and Richard have to share \$360 in the ratio 5 : 4.
This means that for each \$5 Franco receives, Richard receives \$4.

Dividing a quantity in a given ratio

- Step 1:** Find the total number of parts.
(Add the numbers in the ratio.)
- Step 2:** Find the fraction of the quantity for each share.

EXAMPLE

Divide \$360 in the ratio 5 : 4.

Step 1: The total number of parts is $5 + 4 = 9$

Step 2: The two shares are $\frac{5}{9} \times 360$ and $\frac{4}{9} \times 360$
= \$200 and \$160

1 Find the total number of parts in these ratios.

- a** 2 : 3 _____
- b** 3 : 1 _____
- c** 4 : 3 _____
- d** 3 : 5 _____
- e** 4 : 5 _____
- f** 7 : 4 _____

2 Complete these ratio divisions.

- a** Divide \$24 in the ratio 2 : 1.

b Divide \$32 in the ratio 1 : 3.

- c** Divide \$50 in the ratio 2 : 3.

d Divide \$48 in the ratio 3 : 5.

- e** Divide \$42 in the ratio 4 : 3.

f Divide \$80 in the ratio 3 : 2.

- g** Divide \$55 in the ratio 4 : 7.

h Divide \$175 in the ratio 5 : 2.

3.7 Dividing in a ratio

CONTINUED

i Divide 120 cm in the ratio 1 : 4.

k Divide 150 m in the ratio 7 : 8.

m Divide 250 km in the ratio 3 : 2.

j Divide 500 mL in the ratio 2 : 3.

l Divide 180 kg in the ratio 5 : 7.

n Divide 720 mm in the ratio 4 : 5.

Answer the following questions in your workbook.

- 3 Divide 1 kg (1 000 grams) in the ratio 3 : 2. Answer in grams.
- 4 Divide 1 hour (60 minutes) in the ratio 5 : 7. Answer in minutes.
- 5 Stephanie divided \$48 in the ratio 2 : 1. What is the **smaller** share?
- 6 Dylan divided \$200 in the ratio 2 : 3. What is the **larger** share?
- 7 What is the smaller share when \$70 is divided in the ratio 3 : 7?
- 8 There are more girls than boys in Year 8 at Woodport Secondary College. The ratio of girls to boys is 7 : 5. The total number of students in Year 8 is 204. How many girls are in Year 8?
- 9 The total area of a house is 160 square metres. The ratio of the living area to the bedroom area is 5 : 3. What is the living area in this house?
- 10 A sporting club received a grant of \$20 000. The grant was shared between new equipment and team uniforms in the ratio 3 : 5. How much was spent on team uniforms?
- 11 Two stunt riders are to compete for \$3 500 in prize money. The prize money will be divided in the ratio 5 : 2. The winner will receive the larger share. How much will the loser receive?



3.8 Dividing into 3 parts

EXAMPLE

Divide \$720 in the ratio 1 : 2 : 3.

Step 1: The total number of parts is $1 + 2 + 3 = 6$

Step 2: The three shares are $\frac{1}{6} \times 720$, $\frac{2}{6} \times 720$ and $\frac{3}{6} \times 720$
 $= \$120, \240 and $\$360$

1 Divide \$48 in the ratio 1 : 2 : 3.

2 Divide \$72 in the ratio 1 : 3 : 5.

3 Divide \$96 in the ratio 4 : 3 : 1.

4 Divide \$200 in the ratio 2 : 3 : 5.

5 Divide \$225 in the ratio 2 : 4 : 3.

6 Divide \$280 in the ratio 2 : 2 : 3.

Answer the following questions in your workbook.

7 Divide \$440 in the ratio 3 : 1 : 4.

8 Divide \$580 in the ratio 5 : 8 : 7.

9 Sophie divided \$550 in the ratio 2 : 3 : 5.
What was the **smallest** share?

10 Ahmad divided \$420 in the ratio 3 : 4 : 5.
What was the **largest** share?



Teacher's notes

OBJECTIVES

- Recognise and name common polygons.
- Classify triangles by the types of angles they contain and/or the number of equal sides.
- Recognise and name types of quadrilaterals.
- Recognise line symmetry in plane shapes and show the axes of symmetry on a diagram.

PRIOR KNOWLEDGE

Students should understand the meaning of the term **plane shape** and should be familiar with triangles and common quadrilaterals.

BACKGROUND

These activity sheets focus on consolidating students' ability to recognise different polygons, especially triangles and quadrilaterals. They are a useful introduction to the basic properties of these shapes. Students need to be able to recognise shapes presented in different orientations in order to develop flexible mental images.

STARTER ACTIVITY

Work with students to create a class poster showing the plane shapes they know, and label each one with its name and properties.

ACTIVITY SHEETS

4.1 Polygons A

This activity sheet introduces the term **polygon** and the names of common polygons. Draw students' attention to the definition – a polygon has straight sides that do not cross over. Encourage correct spelling of polygon names.

Answers

- 1 **a** pentagon **b** triangle **c** hexagon
d quadrilateral **e** octagon **f** quadrilateral
g heptagon **h** quadrilateral

- 2 **a** 5 **b** 6 **c** 4 **d** 10 **e** 8 **f** 12

- 3 **a, b, d** and **g** are polygons

4.2 Polygons B

This activity sheet develops students' knowledge of polygons and introduces regular polygons and convex polygons.

Answers

- 1 **a** Yes **b** No **c** No **d** Yes **e** No **f** Yes
g Yes **h** Yes

- 2 **a** Yes **b** No **c** Yes **d** No **e** No **f** Yes
g No **h** No

4.3 Triangles

This activity sheet looks at identifying different types of triangles. The first page consists of a cloze activity where students complete the descriptions and notes themselves. The second page provides practice at identifying types of triangles. The two pages may be used together or independently.

Answers

- 1 **a** triangle, acute, less **b** one, equal, right
c angle, more, obtuse **d** three, same
e two, length, size **f** sides, different

- 2 **a** right-angled **b** obtuse-angled **c** acute-angled
d acute-angled **e** obtuse-angled **f** right-angled
g acute-angled **h** obtuse-angled

- 3 **a** isosceles **b** equilateral **c** scalene **d** isosceles
e equilateral **f** isosceles **g** scalene **h** isosceles

- 4 **b** acute-angled isosceles **c** right-angled scalene
d obtuse-angled scalene **e** right-angled isosceles
f acute-angled scalene **g** obtuse-angled isosceles

4.4 Quadrilaterals

This activity sheet introduces the six special quadrilaterals. The first page provides a cloze activity where students complete descriptions of a trapezium, a parallelogram, a rectangle, a rhombus, a square and a kite. The second page provides practice in identifying types of quadrilaterals and activities in building quadrilaterals using triangles. The two pages may be used together or independently.

Answers

- 1 **a** pair, parallel **b** two, opposite **c** angle, right
d four, length **e** sides, each **f** adjacent, same

- 2 **a** square **b** rectangle **c** parallelogram
d rhombus **e** rectangle **f** kite **g** trapezium
h parallelogram **i** rhombus **j** trapezium
k square **l** kite

4.5 Line symmetry

This activity sheet introduces an important concept and property of many plane shapes. It may be useful to provide enlarged copies of the diagrams on the bottom half of this activity sheet as some students will find it easier to investigate symmetry by cutting out the shapes first. Paper folding is a quick and simple means of establishing properties of plane shapes. You could extend this activity by investigating other properties of these shapes.

Answers

- 1 1 2 4 3 3 4 2 5 5 6 0 7 1 8 6 9 0

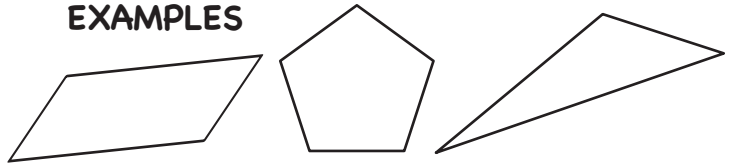
EXTENSION IDEAS

Conduct a class quiz where students have to identify a shape from a description of its features.

4.1 Polygons A

A **polygon** is a plane shape bounded by straight sides which do not cross over.

EXAMPLES



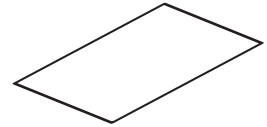
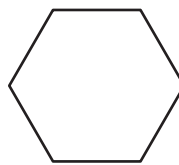
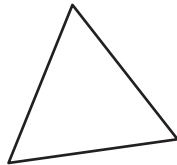
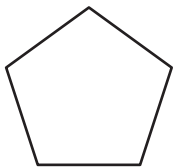
Polygons are named according to the number of sides they have.

- 3 sides: a triangle
- 6 sides: a hexagon
- 9 sides: a nonagon
- 4 sides: a quadrilateral
- 7 sides: a heptagon
- 10 sides: a decagon
- 5 sides: a pentagon
- 8 sides: an octagon
- 12 sides: a dodecagon

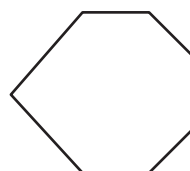
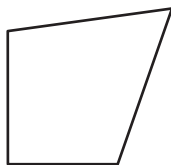
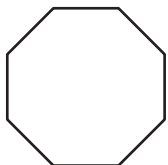
A **regular polygon** has equal sides and equal angles.

1 Name each of the following polygons.

a _____ **b** _____ **c** _____ **d** _____



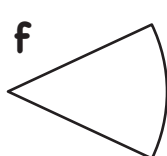
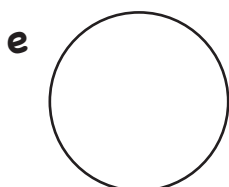
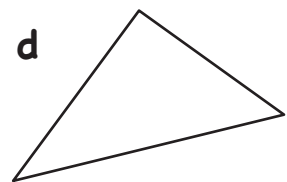
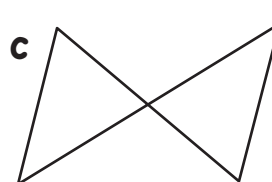
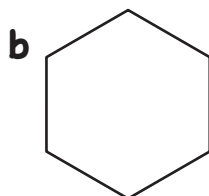
e _____ **f** _____ **g** _____ **h** _____



2 How many sides has:

- a** a pentagon? _____ **b** a hexagon? _____ **c** a quadrilateral? _____
d a decagon? _____ **e** an octagon? _____ **f** a dodecagon? _____

3 Which of the following shapes are polygons? _____

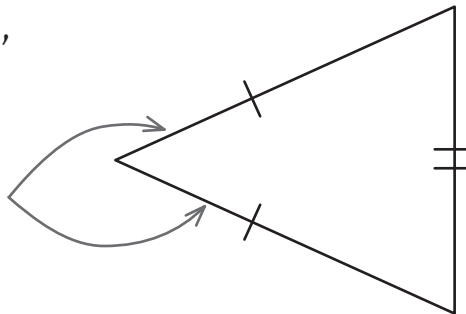


4.2 Polygons B

A **regular polygon** has equal sides and equal angles.

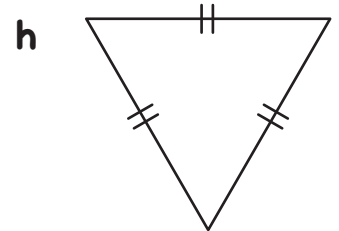
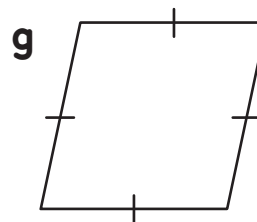
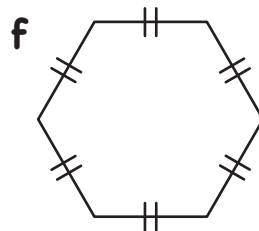
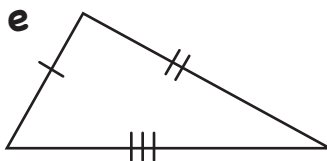
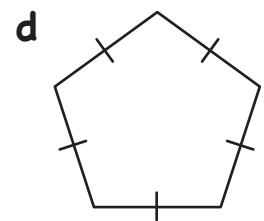
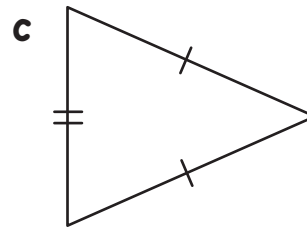
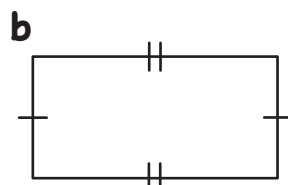
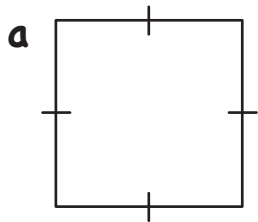
If two sides are the same length, we show this by using the same marking on the diagram.

These two sides are the same length.

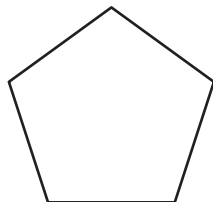


The length of this side is different.

1 Are these polygons regular? Label each one with **Y** (yes) or **N** (no).



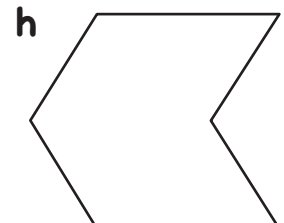
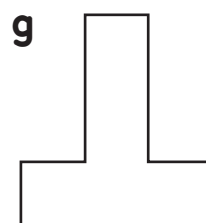
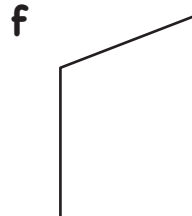
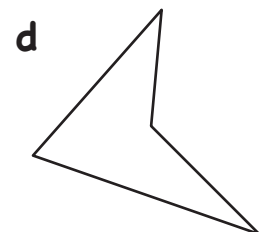
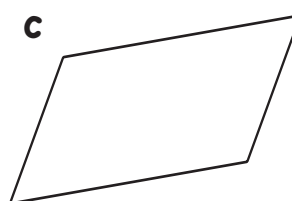
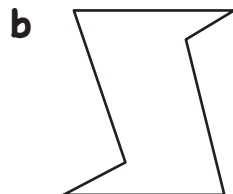
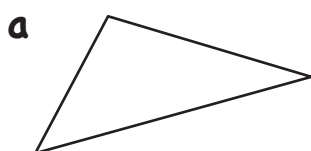
Convex polygons have vertices (corners) that point **outwards**.



Non-convex (or concave) polygons have one or more vertices that point or cave **inwards**.



2 Are these polygons convex? Label each one with **Y** (yes) or **N** (no).



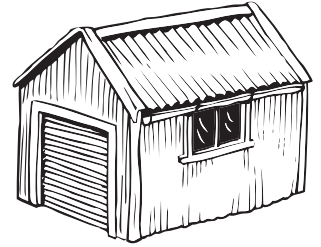
4.3 Triangles

A triangle is an important shape which is used in design, packaging, building and construction.

A **triangle** is a plane shape bounded by 3 straight sides.

A triangle can be classified in two ways:

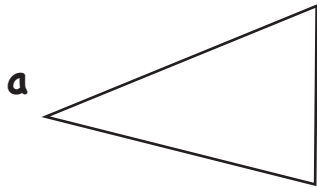
- by the **types of angles** it contains – acute, right or obtuse.
- by the **numbers of equal sides** – equilateral, isosceles or scalene.



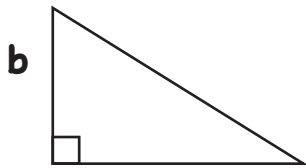
- 1 Use the word bank to complete the sentences. Each word can be used only once.

WORD BANK

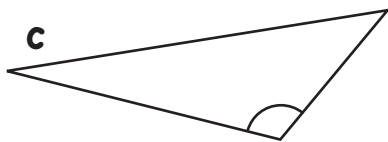
- | | | | |
|-------------|----------|------------|---------|
| • acute | • length | • one | • sides |
| • angle | • less | • right | • size |
| • different | • more | • triangle | • three |
| • equal | • obtuse | • same | • two |



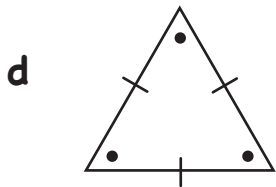
An **acute-angled** _____ has three _____ angles. Each angle measures _____ than 90° .



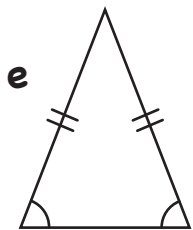
A **right-angled** triangle has _____ of its angles _____ to 90° . One angle is a _____ angle.



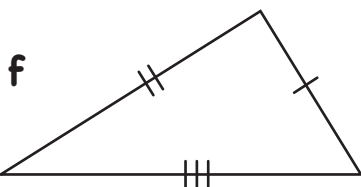
An **obtuse-angled** triangle has one _____ measuring _____ than 90° . One angle is an _____ angle.



In an **equilateral** triangle, the _____ sides are the same length. All three angles are the _____ size.



In an **isosceles** triangle, _____ sides are the same _____. Two angles are the same _____. The equal angles are opposite the equal sides.



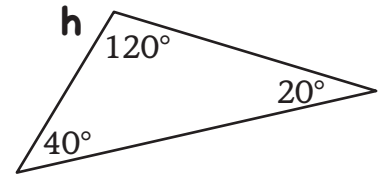
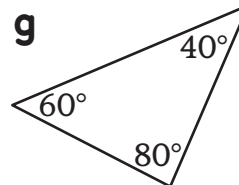
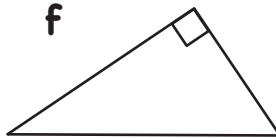
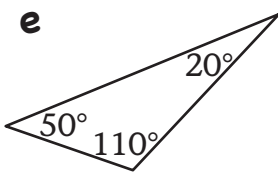
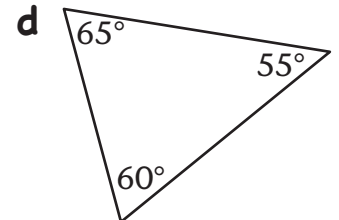
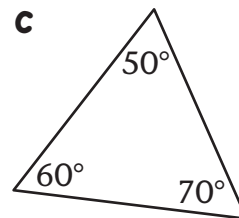
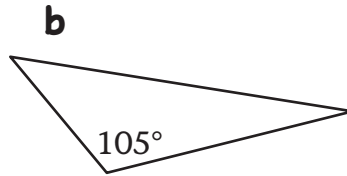
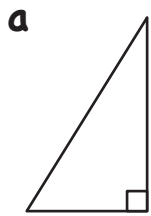
In a **scalene** triangle, the three _____ are different lengths. The three angles are _____ sizes.

4.3 Triangles

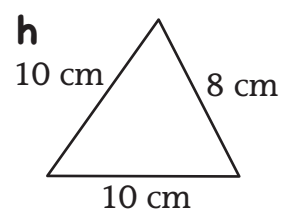
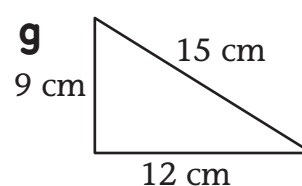
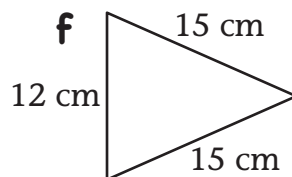
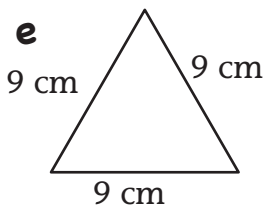
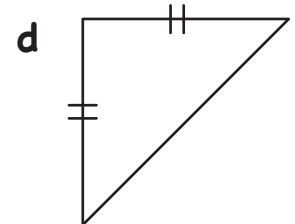
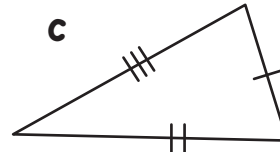
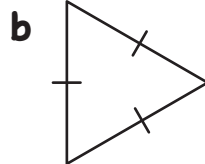
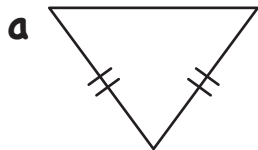
CONTINUED

Answer these questions in your workbook.

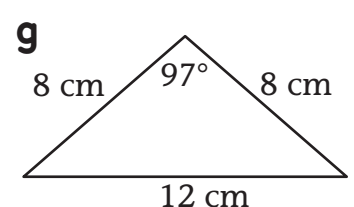
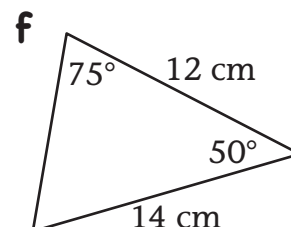
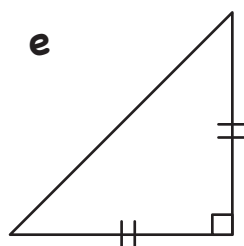
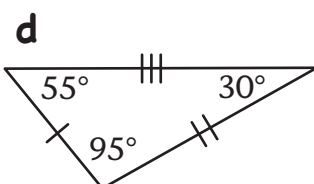
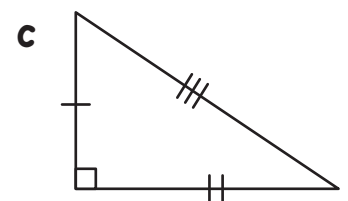
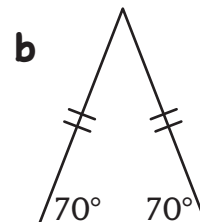
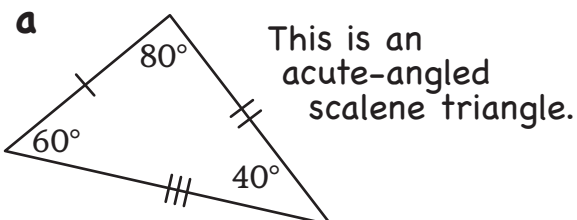
- 2 What type of triangle is each of the following – acute-angled, right-angled or obtuse-angled?



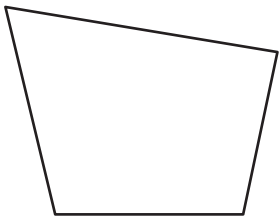
- 3 What type of triangle is each of the following – equilateral, isosceles or scalene?



- 4 Classify each triangle using **two** methods – types of angles **and** lengths of sides.



4.4 Quadrilaterals



Quadrilaterals are plane shapes bounded by four straight sides. There are six special quadrilaterals. They are identified by the number of pairs of parallel sides and the number of sides that are the same length. They are listed in the table below.

In the diagrams, arrows are used to indicate parallel lines.

If two sides are the same length, we show this using the same marking (not arrows – these are used only for parallel lines).

- Use the word bank to complete the sentences in the table. Each word may be used only once.

WORD BANK

- adjacent
- angle
- each
- four
- length
- opposite
- pair
- parallel
- right
- same
- sides
- two

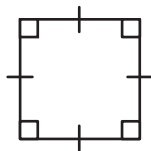
Type of quadrilateral	Diagram	Definition
a Trapezium		A quadrilateral with one _____ of opposite sides _____.
b Parallelogram		A quadrilateral with _____ pairs of _____ sides parallel.
c Rectangle		A quadrilateral in which each _____ measures 90° . It has four _____ angles.
d Rhombus (or diamond)		A quadrilateral in which all _____ sides are the same _____.
e Square		A quadrilateral with all four _____ the same length and _____ angle measuring 90° .
f Kite		A quadrilateral in which two pairs of _____ sides are the _____ length.

4.4 Quadrilaterals

CONTINUED

2 State the type of quadrilateral in each diagram.

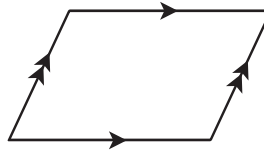
a



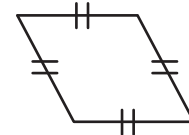
b



c



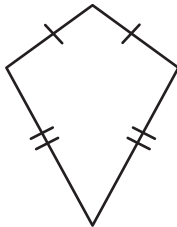
d



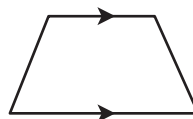
e



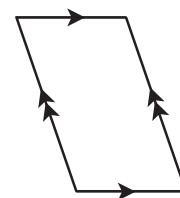
f



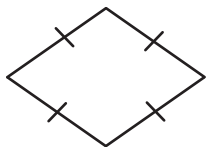
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h



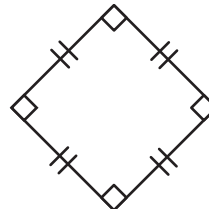
i



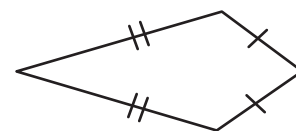
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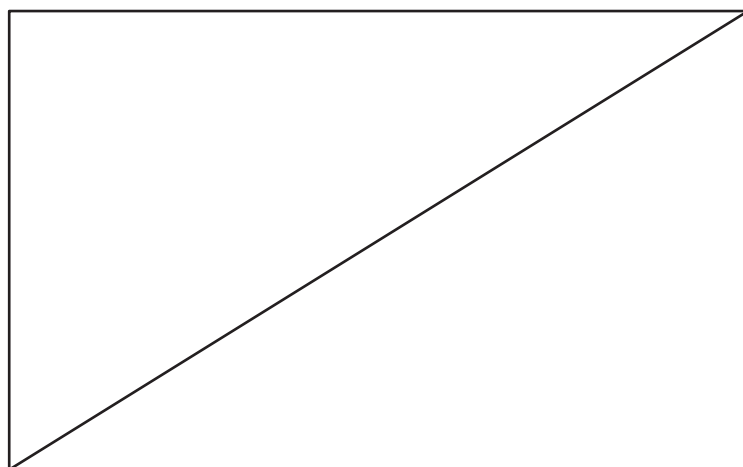
k



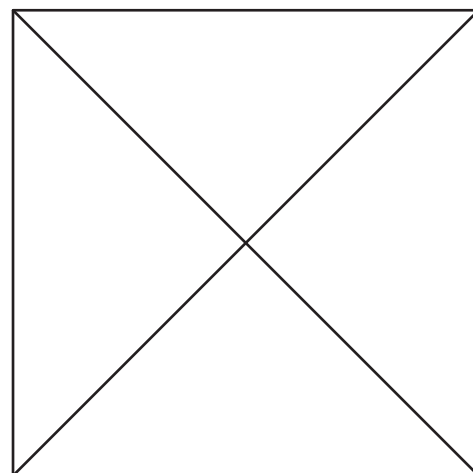
l



- 3 Look at the rectangle below. One diagonal has been drawn. Cut out the rectangle and cut along the diagonal. You now have 2 triangles which are exactly the same. Arrange these 2 triangles to form:
- a parallelogram
 - a kite.

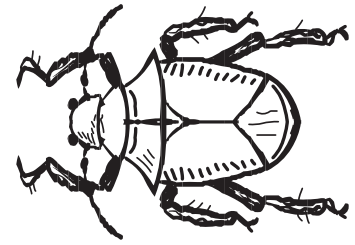


- 4 Look at the square below. Both diagonals have been drawn. Cut out the square and cut along both diagonals. You now have 4 triangles which are exactly the same. Arrange these 4 triangles to form:
- a rectangle
 - a trapezium
 - a parallelogram.



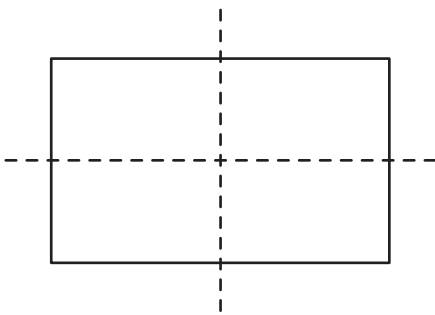
4.5 Line symmetry

Many designs and buildings are symmetrical. They look well balanced and attractive to us. There are many examples of symmetry in nature, including beetles and butterflies.



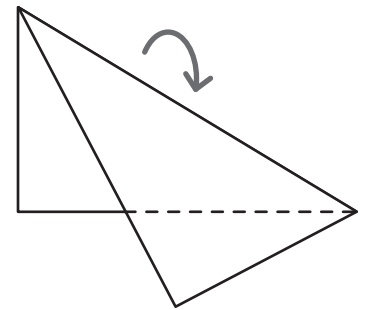
A plane shape has **line symmetry** if it can be folded along a line or reflected in a mirror so that the two halves match exactly. The line of the fold is called an **axis of symmetry**.

Some shapes have more than one axis of symmetry. For example, a rectangle can be folded in half 2 ways, as you can see in the diagram on the left below.



We use dotted lines to show each axis of symmetry.

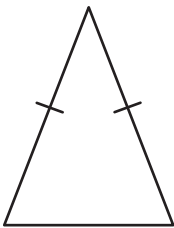
However, if you try folding a rectangle along its diagonal, the two halves do not match exactly. This is **not** an axis of symmetry.



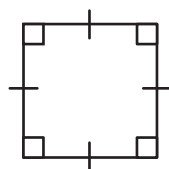
Investigate the line symmetry of the following shapes.

- Use dotted lines to show each axis of symmetry on the diagrams.
- State the number of axes of symmetry for each shape.

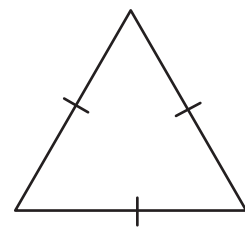
1 isosceles triangle



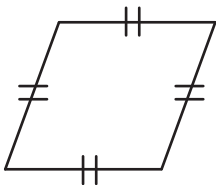
2 square



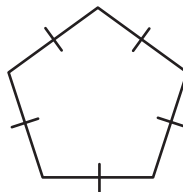
3 equilateral triangle



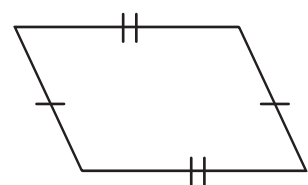
4 rhombus



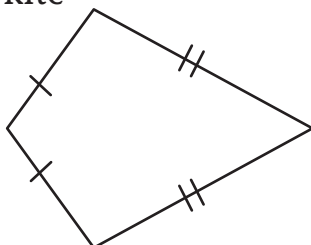
5 regular pentagon



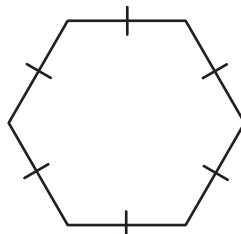
6 parallelogram



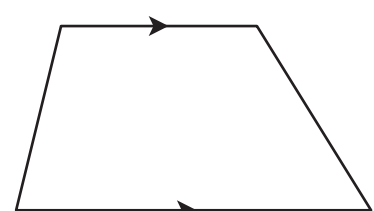
7 kite



8 regular hexagon



9 trapezium



Teacher's notes

OBJECTIVES

- Use a ruler and a set square to draw parallel lines.
- Use a ruler and a set square to draw perpendicular lines.
- Use a protractor to measure the size of an angle.
- Use a protractor to draw an angle of a given size.
- Use a pair of compasses to draw a circle.

PRIOR KNOWLEDGE

Before using these activity sheets, students should understand the meaning of the words **line, parallel, perpendicular, angle, vertex, clockwise, anticlockwise, circle and arc**.

BACKGROUND

While mathematical templates and computer software packages are useful tools for drawing and investigating geometrical figures, many students will need to be able to use geometric instruments in their lives after school. These activity sheets focus on developing students' drawing skills using geometric instruments. They are a useful introduction to any unit of work dealing with geometric constructions.

Note: Metal-edged wooden rulers are **not** recommended for drawing work, as the metal edge is not always straight.

STARTER ACTIVITY

Work with students to draw and label diagrams to illustrate the meanings of the terms *line, parallel, perpendicular, angle, vertex, clockwise, anticlockwise, circle and arc*.

ACTIVITY SHEETS

5.1 Parallel lines

This activity sheet shows students how to use a ruler and a set square to draw a pair of parallel lines by following step-by-step instructions. Students will need a ruler and a set square to complete this activity sheet. Students complete question 2 in their workbooks.

5.2 Perpendicular lines

This activity sheet shows students how to use a ruler and a set square to draw a pair of perpendicular lines by following step-by-step instructions. Students will need a ruler and a set square to complete this activity sheet.

Answers

1 a Yes b Yes c No

5.3 Using a protractor

This activity sheet shows students how to use a protractor to measure an angle. Students do **not** need a protractor to complete this activity sheet.

Answers

1 50° 2 130° 3 40° 4 65°

Note: On activity sheets 5.3, 5.4 and 5.5, the diagrams show a 180° protractor with an outer anticlockwise scale. All protractors are **not** identical. Teachers should check that students have a similar protractor before using these activity sheets.

5.4 Measuring angles

This activity sheet provides practice at measuring the size of an angle. Students will need a protractor to complete this activity sheet. It may be necessary for students to extend the arms of the angles before they measure them, depending on the size of their protractors.

Answers

1 60° 2 45° 3 110° 4 50° 5 35° 6 115°
7 55° 8 125°

5.5 Drawing angles

This activity sheet shows students how to use a protractor to draw an angle of a given size. Encourage students to follow steps 2 to 5 using the base line AB provided. Students will need a ruler and a protractor to complete this activity sheet.

5.6 Using compasses

This activity sheet provides tips on using compasses and practice at drawing circles. Students will need a ruler and a pair of compasses, and they need to complete these activities in their workbooks or on a sheet of drawing paper.

Answers

3 e a square

EXTENSION IDEAS

Have students use the skills they have learnt to create an abstract design according to given combinations of lines, angles and circles, or a stylised version of a real-life object.

5.1 Parallel lines

DEFINITIONS

A **line** extends indefinitely in each direction.

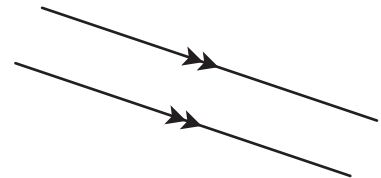
An arrow is often placed on each end to show this.



Lines that never meet, no matter how far they are extended, are called **parallel lines**.

Two parallel lines are always the same distance apart.

We put small arrows on the lines to indicate that they are parallel.



You can draw a pair of parallel lines using a ruler and a set square.

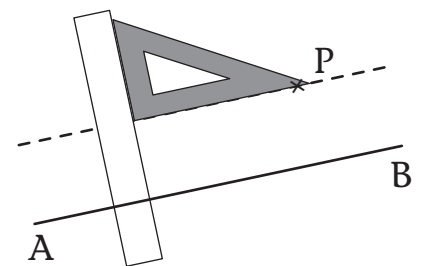
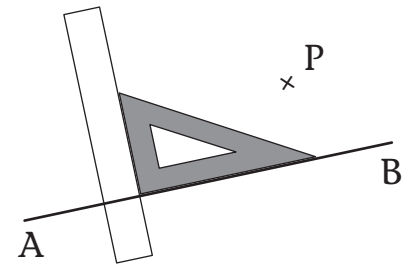
Draw a line through P parallel to the line AB.

Step 1: Place the edge of your set square along the line, as shown.

Step 2: Place your ruler against the other side of the set square.

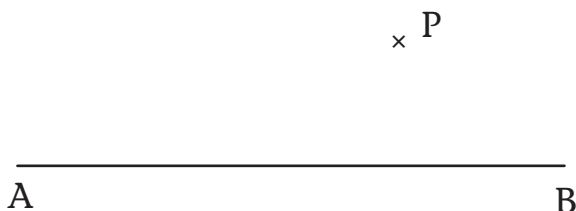
Step 3: Hold the ruler down firmly. Slide the set square along the ruler to the point P.

Step 4: Draw a line along the same edge of the set square, through P.

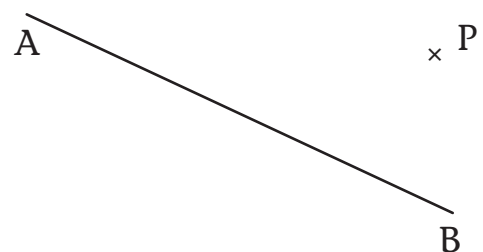


1 Use a set square and a ruler to draw a line through P parallel to the line AB in each diagram.

a



b

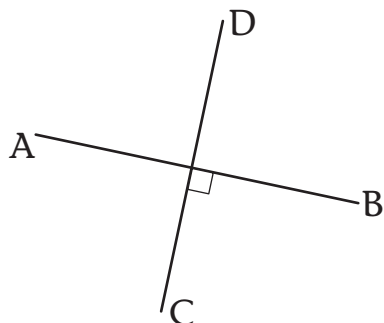


Answer this question in your workbook.

2 Draw a line PQ. Mark a point A, not on this line. Use a ruler and a set square to draw a line through A, parallel to PQ.

5.2 Perpendicular lines

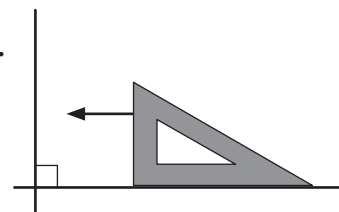
Perpendicular lines meet at right angles (90°) to each other.



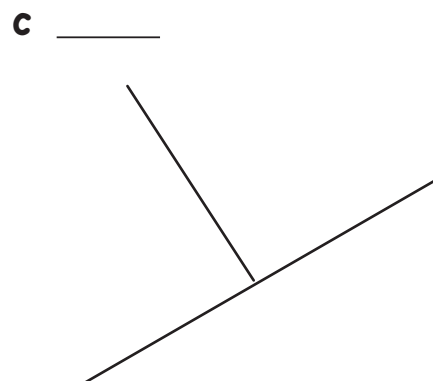
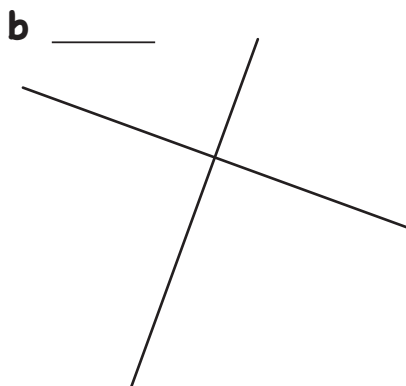
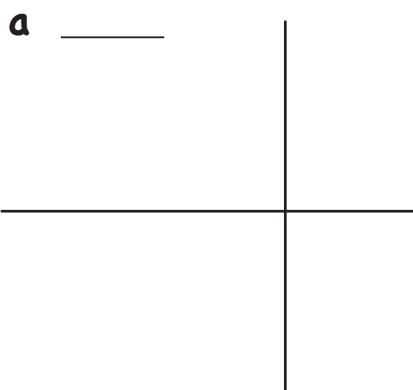
In the diagram at the left, AB is perpendicular to CD ($AB \perp CD$).

We draw a small square in the angle to indicate it is a right angle.

A set square provides an easy way to check right angles.

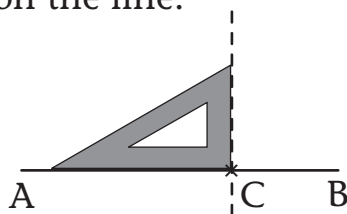


1 Is each pair of lines perpendicular? Use a set square to check.

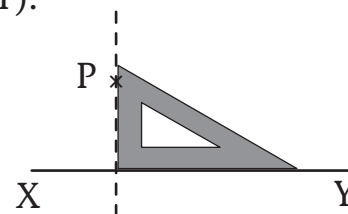


You can use a set square to construct a pair of perpendicular lines.

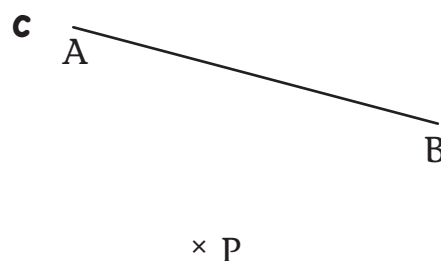
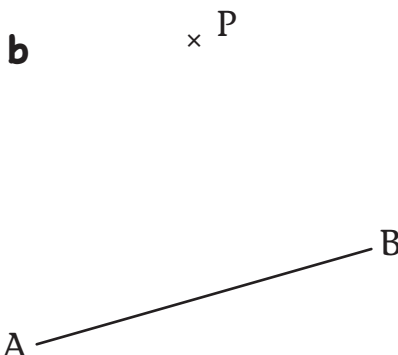
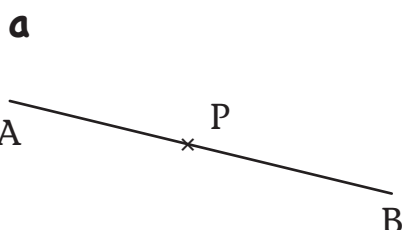
- Constructing a line perpendicular to AB at a point C on the line:



- Constructing a line perpendicular to XY to pass through a point P (not on XY):



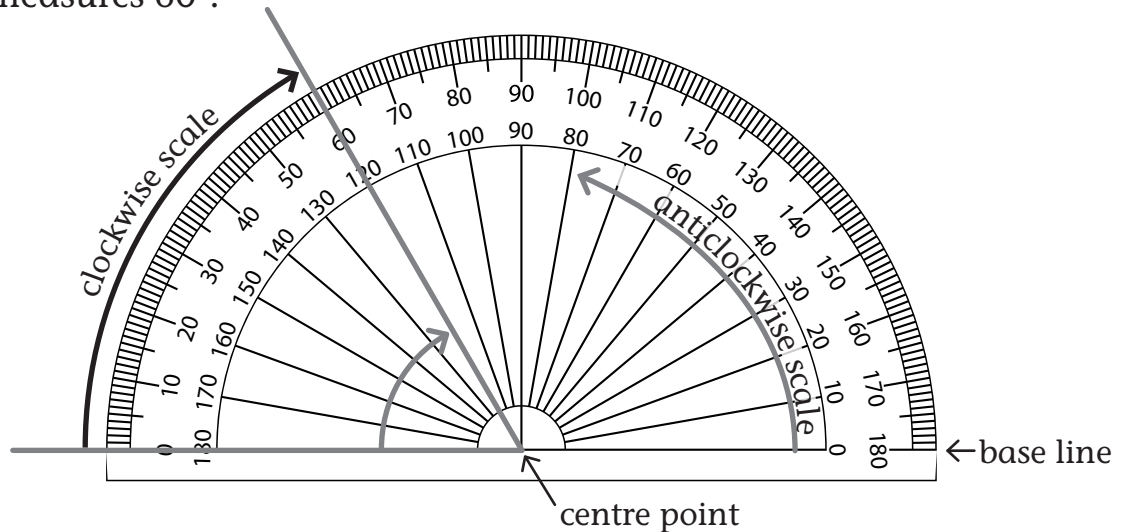
2 Use a set square to draw a line through P perpendicular to AB in each diagram.



5.3 Using a protractor

The size of an angle is the amount of turning between two lines which meet at a point. Angles are measured in degrees, using a protractor.

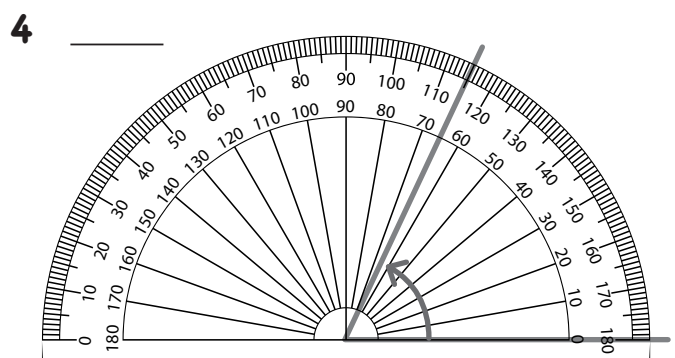
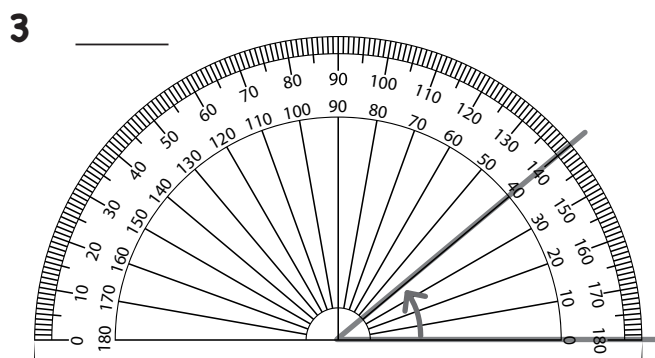
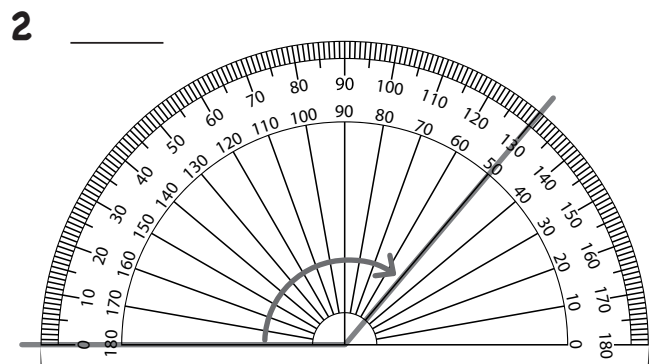
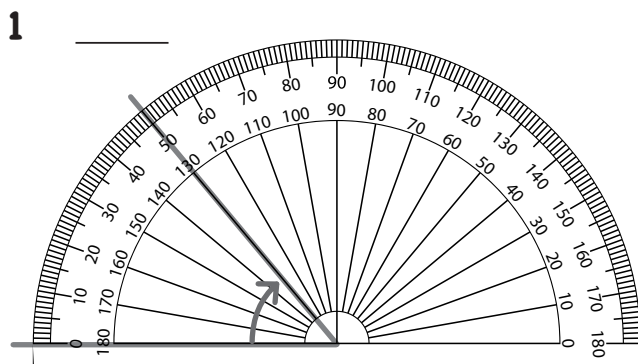
This angle measures 60° .



Most protractors have two scales: a **clockwise scale** (left to right) and an **anticlockwise scale** (right to left).

When you use a protractor, make sure that the centre point is at the vertex of the angle. The base line should be along one of the arms of the angle.

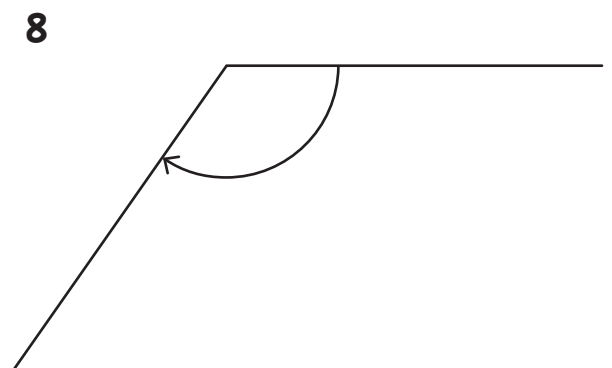
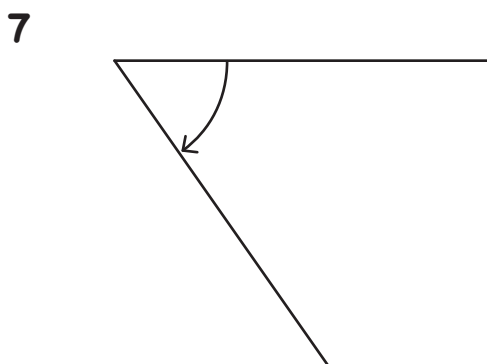
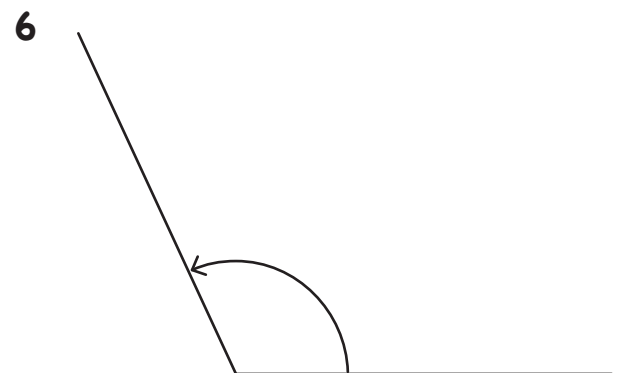
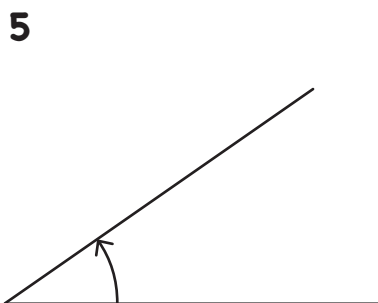
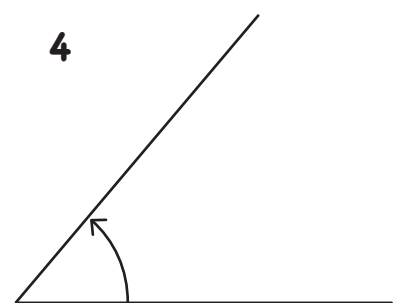
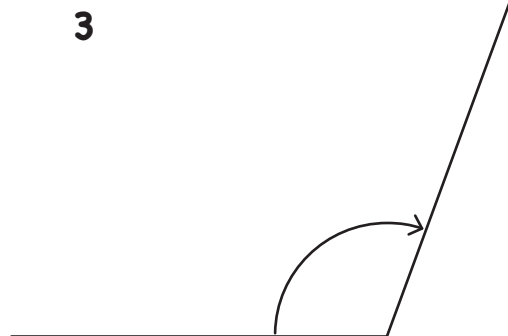
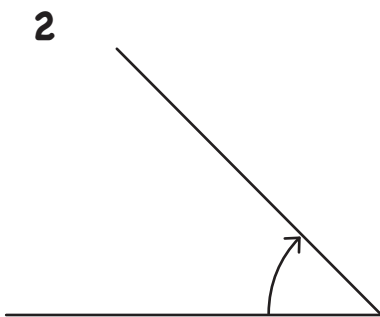
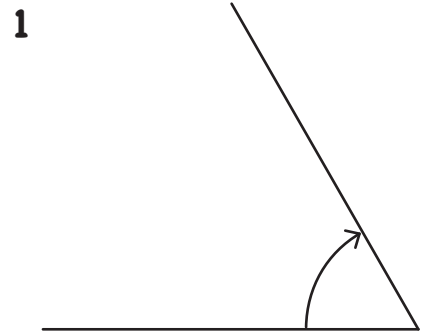
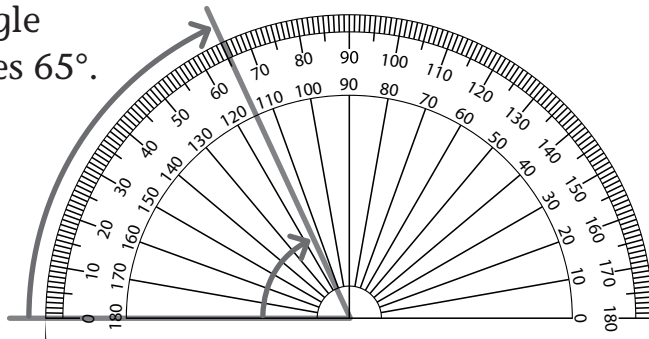
In each diagram below, a protractor has been placed over an angle. Read the scale on the protractor to find the size of each angle.



5.4 Measuring angles

Use a protractor to measure the size of these angles.

This angle measures 65° .



5.5 Drawing angles

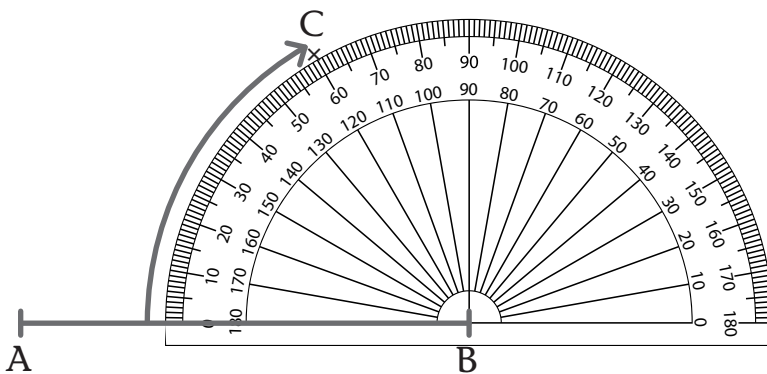
EXAMPLE

Use the **clockwise** (outside) scale of your protractor to draw an angle measuring 60° .

Step 1: Draw a line 6 cm long. Label the end points A and B, as shown at the right.

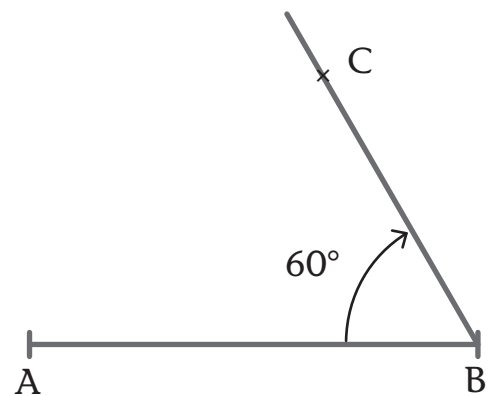
Step 2: Place the base line of your protractor along this line.

Step 3: Move the centre point to the right-hand end (B), as shown below.



Step 4: Follow the clockwise (outside) scale of your protractor around to 60. Mark this point C.

Step 5: Draw a line from C to B, as shown at the right. Angle ABC measures 60° .



Construct the following angles in your workbook.

1 Use the **clockwise** scale of your protractor to draw angles measuring:

- a** 40° . **b** 70° . **c** 65° . **d** 130° . **e** 115° .

When you use the **anticlockwise** scale of your protractor, place the centre point on the left-hand end point (A). Join point C to A.

2 Use the **anticlockwise** scale of your protractor to draw angles measuring:

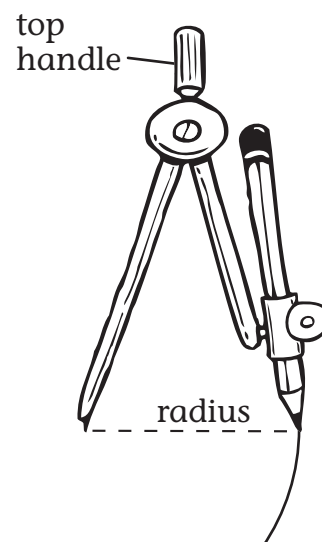
- a** 50° . **b** 30° . **c** 45° . **d** 120° . **e** 165° .

5.6 Using compasses

Compasses are very handy instruments.
They can be used to draw circles or arcs.
They are used in many geometric constructions.

Technical tips

- Line up the metal point and the tip of your pencil before using the compasses.
- Be sure the legs of your compasses are securely positioned before you start to draw.
- Place something between the sheet of paper and the desk, so that the metal tip does not slide.
- Use the scale on your ruler to open the compasses to the correct radius.
- When you are drawing a circle or an arc, hold your compasses by the top handle, not by the two legs.



Complete the following constructions in your workbook.

1 Use a pair of compasses to draw 3 circles.

a radius 5 cm

b radius 8 cm

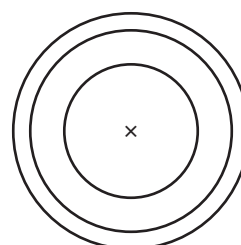
c radius 65 mm

2 Use a pair of compasses and the **same centre** to draw 3 circles with radius:

a 4 cm.

b 6 cm.

c 7 cm.



These are called **concentric circles** because they have the same centre.

3 Refer to the diagram at the right to help you complete the following steps.

a Construct a circle with radius 45 mm.

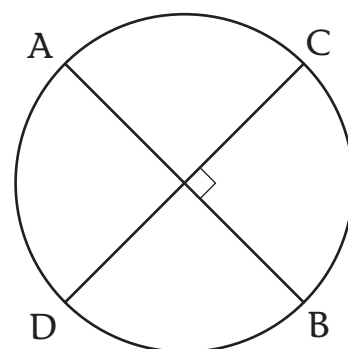
b Draw a diameter AB.

(A diameter is a line drawn from one side of a circle to the other, passing through the centre.)

c Use a set square to draw a second diameter CD, perpendicular to AB.

d Join the points A, C, B and D.

e What type of figure is ACBD?



Teacher's notes

OBJECTIVES

- Use a ruler and a set square to construct a rectangle or a square.
- Use a ruler and a protractor to construct a parallelogram.
- Use a ruler and a pair of compasses to construct a triangle.
- Construct a regular polygon inside a circle.

PRIOR KNOWLEDGE

These activity sheets assume that students know how to use geometric instruments for measurement and drawing. Before using the activity sheets, students should be able to identify **triangles, squares, rectangles and parallelograms**. They should also understand the meaning of the words: **construct, point, line, angle, arc, circle, right angle and polygon**.

BACKGROUND

These activity sheets can be used to develop students' knowledge and skills in constructing common plane shapes. They may be used independently or as a unit of work.

Note: Metal-edged wooden rulers are **not** recommended for drawing work, as the metal edge is not always straight.

STARTER ACTIVITY

Work with the students to draw diagrams of triangles, squares, rectangles and parallelograms, and to label each one with its name and properties. Then review the terms *construct, point, line, angle, arc, circle, right angle and polygon*. Have the students draw diagrams to show their understanding of each term.

ACTIVITY SHEETS

For the activity sheets in this topic, students need to complete the tasks in their workbooks or on a sheet of drawing paper.

6.1 Constructing rectangles and squares

This activity sheet provides step-by-step instructions for constructing a rectangle. Students will need a ruler and a set square. Encourage students to follow steps 2 to 4 using the base line AB provided below step 1 on the activity sheet.

6.2 Constructing parallelograms

This activity sheet provides step-by-step instructions for constructing a parallelogram. Students will need a ruler and a protractor. Encourage students to follow steps 2 to 4 using the base line PS provided below step 1 on the activity sheet.

6.3 Constructing triangles

This activity sheet provides step-by-step instructions for constructing a triangle given the lengths of the 3 sides. Students will need a ruler and a pair of compasses. Encourage students to follow steps 2 to 4 using the base line provided below step 1 on the activity sheet.

6.4 Constructing regular polygons

This activity sheet provides step-by-step instructions for constructing a regular hexagon. They are then asked to repeat the steps using a different angle to construct a regular octagon. Students will need a ruler, a protractor and a pair of compasses.

EXTENSION IDEAS

Have students extend the work done in activity sheet 6.4 **Constructing regular polygons** by working out how to construct different polygons such as pentagons or nonagons. (Angle required = $360^\circ \div \text{number of sides}$)

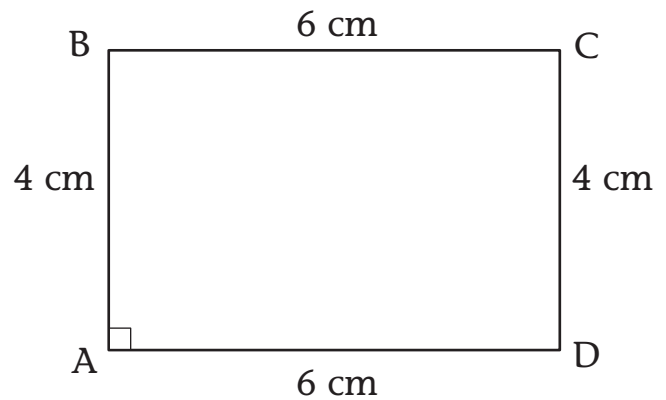
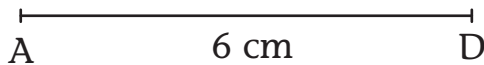
Then students could apply the skills they have learned to experimenting with abstract design and symmetry, or replicating/stylising a real-life construction such as their own home, an interior wall of the classroom or another school building.

6.1 Constructing rectangles and squares

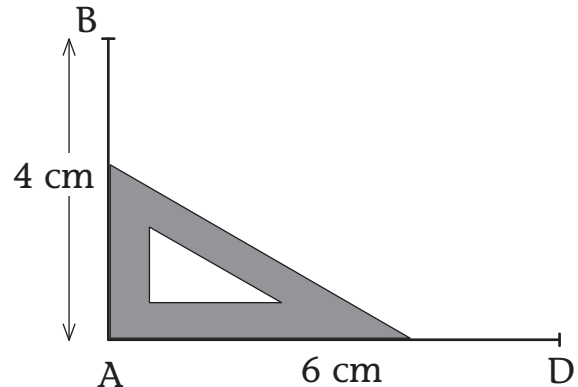
Use a ruler and a set square to construct a rectangle 6 cm long and 4 cm wide by following the steps below.

Remember, all angles in a rectangle are right angles.

Step 1: Use a ruler to draw the base line (AD) 6 cm long, as shown below. Allow space above this line to construct the rectangle.



Step 2: Use a set square to draw AB at right angles to AD, as shown below. Use your ruler to mark the point B, 4 cm from A.

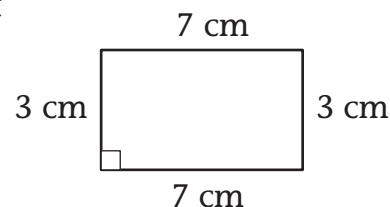


Step 3: Use your set square to draw DC at right angles to AD. Use your ruler to mark the point C, 4 cm from D.

Step 4: Use your ruler to join BC.

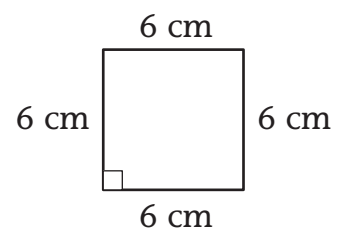
Complete the following constructions in your workbook.

1 Use a ruler and a set square to construct a rectangle 7 cm long and 3 cm wide.



2 Construct a rectangle 65 mm long and 35 mm wide.

3 Use a ruler and a set square to construct a square with sides 6 cm long. Remember, all four sides of a square are the same length.

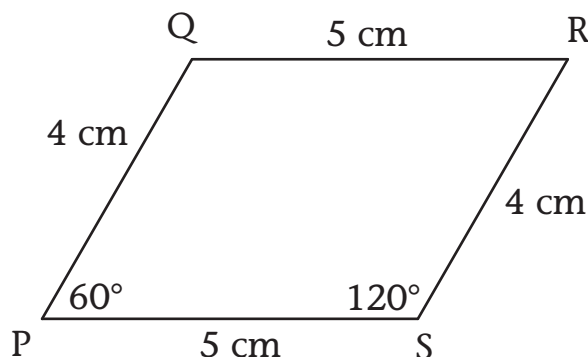


4 Construct a square with sides 45 mm long.

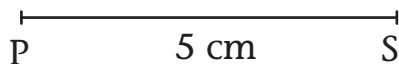
6.2 Constructing parallelograms

Use a ruler and a protractor to construct this parallelogram by following the steps below.

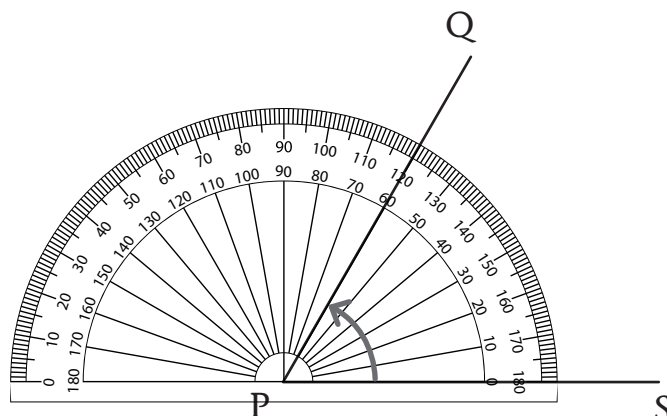
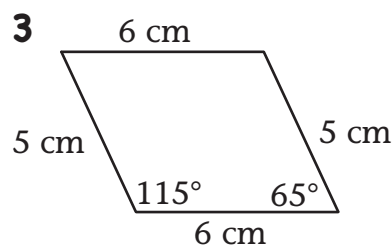
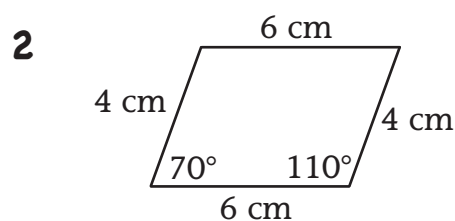
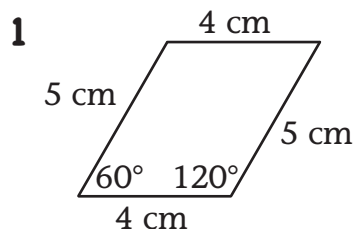
Step 1: Use a ruler to draw the base line (PS) 5 cm long, as shown below. Allow space above this line to construct the parallelogram.



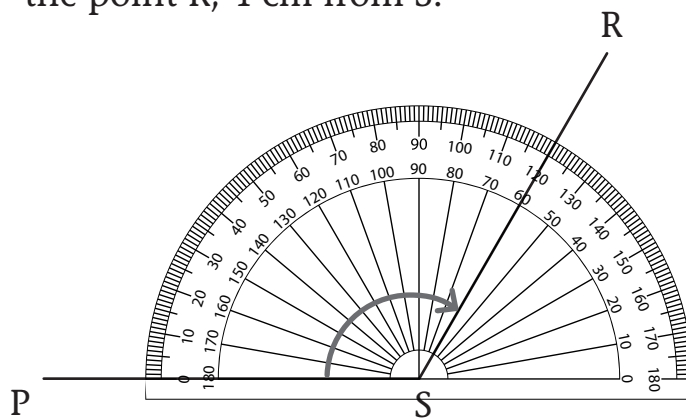
Step 2: Use a protractor to draw angle P measuring 60° . Use your ruler to mark the point Q, 4 cm from P.



Use a ruler and a protractor to construct these parallelograms in your workbook. (These diagrams are **not** the correct size.)



Step 3: Use a protractor to draw angle S measuring 120° . Use your ruler to mark the point R, 4 cm from S.



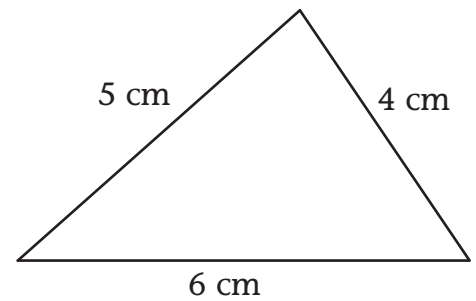
Step 4: Use your ruler to join QR.

6.3 Constructing triangles

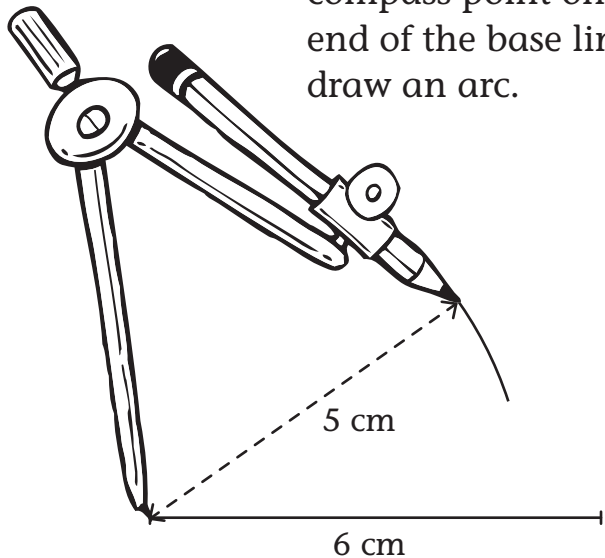
Use a ruler and a pair of compasses to construct a triangle with sides measuring 6 cm, 5 cm and 4 cm.

Follow these steps to construct a triangle when you are told the lengths of the 3 sides.

Step 1: Use your ruler to draw the first side (6 cm) as shown below. Leave space above it to complete the triangle.

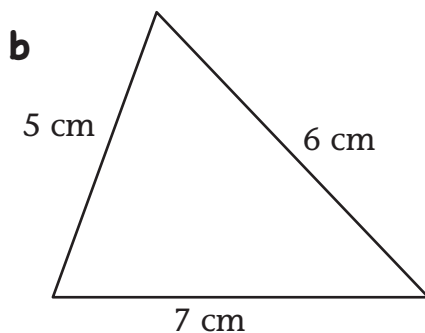
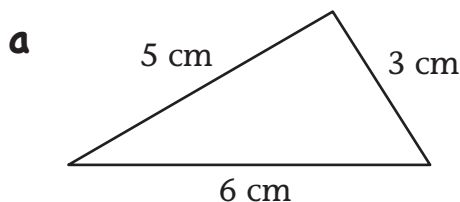


Step 2: Open your compasses to a radius of 5 cm. Place the metal compass point on one end of the base line and draw an arc.



Complete the following constructions in your workbook.

1 Construct these triangles at the correct size.

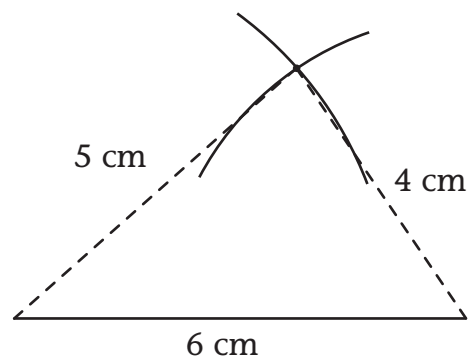


2 Use a ruler and a pair of compasses to construct a triangle with sides measuring:

a 70 mm, 65 mm and 55 mm.

b 85 mm, 60 mm and 65 mm.

Step 3: Open your compasses to a radius of 4 cm. Draw an arc from the other end of the base line. Mark the point where the two arcs cross, as shown in the diagram below.



Step 4: Use your ruler to draw lines from this point to each end of the base line.

6.4 Constructing regular polygons

A **polygon** is a plane shape bounded by straight sides, which do not cross over.

A **regular polygon** has equal sides and equal angles.

A circle is often used as the base for constructing a regular polygon. You can always draw a circle through the vertices of a **regular polygon**.

You need a pair of compasses, a protractor and a ruler for the constructions on this activity sheet.



- Follow the steps below to construct a **regular hexagon** (6 sides) in your workbook.

Step 1: Use a pair of compasses to construct a circle with radius 5 cm. (Alternatively, you could trace around a 360° protractor.)

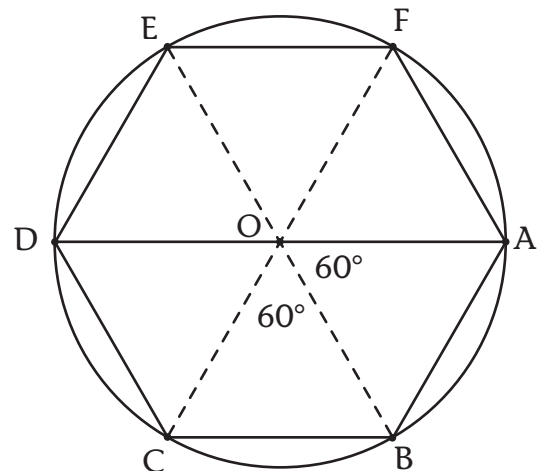
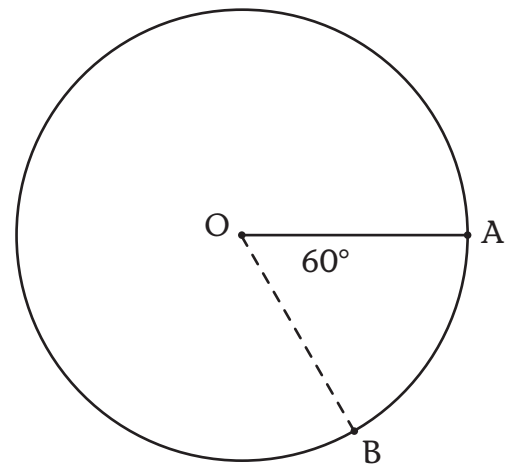
Step 2: Mark the centre of your circle and label it point O.

Step 3: Draw a radius OA.

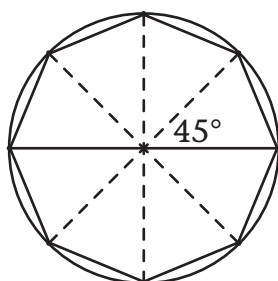
Step 4: Use your protractor to construct an angle BOA measuring 60° .

Step 5: Now construct an angle COB also measuring 60° . Then repeat this process to divide the circle into 6 sectors, each containing an angle of 60° .

Step 6: Join the points A, B, C, D, E and F to form a regular hexagon.



- Repeat the steps in question 1 but use an angle of 45° instead of 60° to construct a **regular octagon** (8 sides).



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